Chapter 2

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Chapter 2 Free Oscillations of Systems with Many Degrees of Freedom

2.1 Introduction

In Chap. 1 we studied oscillations of systems having one or two degrees of freedom. In this chapter we shall study systems having N degrees of freedom, where N can range up to some very large number, which we shall loosely call "infinity."

For a system with N degrees of freedom, there are always exactly N modes (see Prob. 1.17). Each mode has its own frequency ω and its own "shape" given by the amplitude ratios $A:B:C:D:\ldots$ etc., corresponding to the degrees of freedom $a, b, c, d, \ldots$, etc. In each mode, all moving parts go through their equilibrium positions simultaneously; that is, every degree of freedom oscillates in that mode with the same phase constant. Thus there is a single phase constant for the entire mode, which is determined by the initial conditions. Since each degree of freedom oscillates in a given mode with the same frequency ω, each moving part experiences the same return force per unit displacement per unit mass, given by ω^2.

As an example, suppose we have a system with four degrees of freedom a, b, c, d. Then there are four modes. Suppose that in mode 1 the amplitude ratios are

$$A:B:C:D = 1:0:-2:7.$$

Then the motions of a, b, c, and d (if mode 1 is the only excited mode) are given by

$$\psi_a = A_1 \cos(\omega_1 t + \varphi_1), \qquad \psi_b = 0, \qquad \psi_c = -2\psi_a, \qquad \psi_d = 7\psi_a,$$

where A_1 and φ_1 depend on the initial conditions.

If a system contains a very large number of moving parts, and if these parts are distributed within a limited region of space, the average distance between neighboring moving parts becomes very small. As an approximation, one may wish to think of the number of parts as becoming infinite and the distance between neighboring parts as going to zero. One then says that the system behaves as if it were "continuous." Implicit in this point of view is the assumption that the motion of near neighbors is nearly the same. This assumption allows us to describe the vector displacement of all the moving parts in a small neighborhood of a point x, y, z, with a single vector quantity $\psi(x,y,z,t)$. Then the "displacement" $\psi(x,y,z,t)$ is a contin-

uous function of position, x, y, z, and of time t. It replaces the description using the displacements $\psi_a(t)$, $\psi_b(t)$, etc., of the individual parts. We then say we are dealing with *waves*.

Standing waves are normal modes. The modes of a continuous system are called *standing waves*, or *normal modes*, or simply *modes*. According to the discussion above, a truly continuous system has an infinite number of independent moving parts, although they occupy a finite space. There are therefore an infinite number of degrees of freedom, and hence an infinite number of modes. This is not literally true for a real material system. One liter of air does not contain an infinite number of moving parts, but only 2.7×10^{22} molecules, each of which has three degrees of freedom (for motion along x, y, and z directions). Thus a bottle containing 1 liter of air does not have an infinite number of possible vibrational modes of the air, but only 8×10^{22} at most. Anyone who has practiced blowing a bottle or a flute knows that it is not easy to excite more than the first few modes. (We usually distinguish the modes by calling the one with the lowest frequency number 1, the next higher number 2, etc.) In practice we are often concerned only with the first few (or few dozen or few thousand) modes. As we shall see, it turns out that the lowest modes behave as if the system were continuous.

The most general motion of a system can be written as a superposition of all its modes, with the amplitude and phase constant of each mode set by the initial conditions. The appearance of the vibrating system in such a general situation is very complicated, simply because the eye and brain cannot contemplate several things at once without confusion. It is not easy to look at the complete motion and "see" each mode separately when many are present.

Modes of beaded string. We study first the transverse oscillations of beaded strings. By "strings" we shall really mean springs. We will assume that we have linear (i.e., Hooke's law) massless springs connecting point masses M. (In our figures, we will draw the springs as straight lines rather than as helices.)

In Fig. 2.1 we exhibit a sequence of systems of beaded strings. The first system has $N = 1$ (one degree of freedom), the next $N = 2$, etc. In each case, we exhibit without proof the configurations of the normal modes. Later we shall derive the exact configuration and frequency for each mode.

It should already be possible for you to see (assuming the configurations shown are those of the modes) that we have correctly ordered the configurations in order of ascending mode frequency. That is because the strings

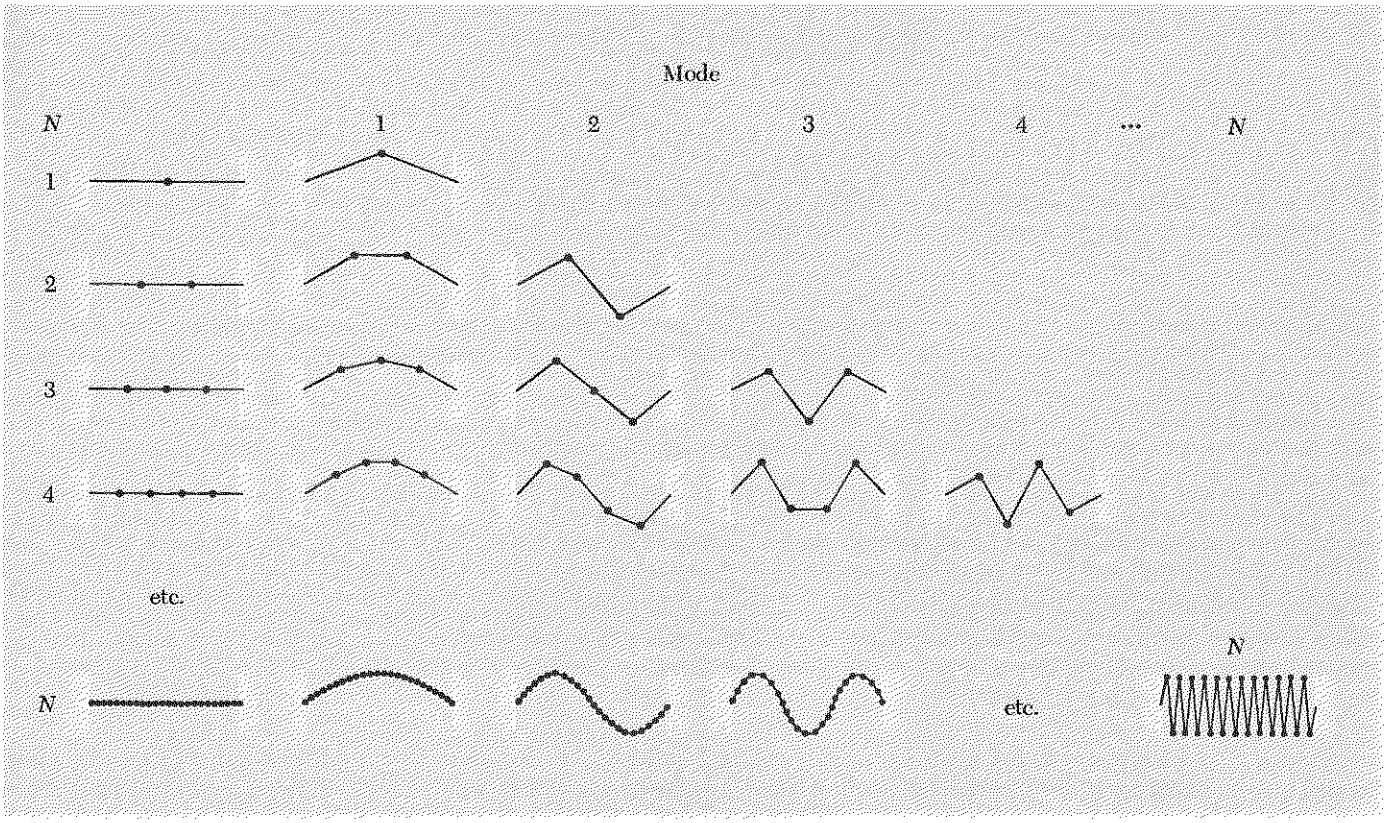


Fig. 2.1 ***Transverse vibrational modes of a beaded string. A string with N beads has N modes. In mode m the string crosses the equilibrium axis m − 1 times and has m half-wavelengths. The highest frequency mode is the "zig-zag" configuration shown.***

make increasingly large angles with the equilibrium axis as we increase the mode number (taking the displacement of a given bead to be the same). Consequently the return force per unit displacement per unit mass for a given bead in a given system increases when we go from one configuration to the next, and therefore so does the mode frequency.

Another thing that is apparent is that our sequence of assumed mode shapes always gives exactly N configurations: the first mode always has zero "nodes" (places where the string crosses the axis, excluding the end points), the second has one node, etc. The highest mode always has the largest possible number of nodes, namely $N - 1$, which is achieved by "zig-zagging" up and down, i.e., crossing the axis once between each two successive masses.

2.2 *Transverse Modes of Continuous String*

We now consider the case where N is huge, say $N = 1{,}000{,}000$ or so. Then for the lowest modes (say the first few thousand), there are a very large number of beads between each node. Thus the displacement varies

slowly from one bead to the next. [We shall *not* consider here the highest modes, since they approach the "zigzag limit," where a description using a continuous function $\psi(x,y,z,t)$ is not possible.] Therefore, in accordance with the remarks above, we shall not describe the instantaneous configuration by the list of displacements $\psi_a(t)$, $\psi_b(t)$, $\psi_c(t)$, $\psi_d(t)$, etc., of each bead. Instead we consider all the particles with *equilibrium* positions in the neighborhood of the point x,y,z (a neighborhood being an infinitesimal cube, if you wish, with edges of length Δx, Δy, and Δz) as having the same instantaneous vector displacement $\psi(x,y,z,t)$:

$$\psi(x,y,z,t) = \hat{\mathbf{x}}\psi_x(x,y,z,t) + \hat{\mathbf{y}}\psi_y(x,y,z,t) + \hat{\mathbf{z}}\psi_z(x,y,z,t), \tag{1}$$

where $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ are unit vectors and ψ_x, ψ_y, and ψ_z are the components of the vector displacement ψ. It is important to realize that x,y,z label the *equilibrium* position of the particles in that neighborhood. Thus x,y,z are not functions of time.

Longitudinal and transverse vibration. Equation (1) is of a much more general form than we need in order to study the vibrations of a string. Suppose that at equilibrium the string is stretched along the z axis. Then the coordinate z is sufficient to label the equilibrium position of each bead (to an accuracy Δz) and Eq. (1) can be written in the simpler form

$$\psi(z,t) = \hat{\mathbf{x}}\psi_x(z,t) + \hat{\mathbf{y}}\psi_y(z,t) + \hat{\mathbf{z}}\psi_z(z,t). \tag{2}$$

Vibrations along the z direction are called *longitudinal* vibrations. Vibrations along the x and y directions are called *transverse* vibrations. At present we wish to consider only the transverse vibrations of the string. Therefore we assume ψ_z is zero:

$$\psi(z,t) = \hat{\mathbf{x}}\psi_x(z,t) + \hat{\mathbf{y}}\psi_y(z,t). \tag{3}$$

Linear polarization. As a further simplification, we assume that the vibrations are entirely along $\hat{\mathbf{x}}$ (i.e., $\psi_y \equiv 0$). The vibrations are then said to be *linearly polarized* along $\hat{\mathbf{x}}$. (In Chap. 8 we shall study general states of polarization.) Now we can drop the unit vector $\hat{\mathbf{x}}$ and the subscript on ψ_x from the notation:

$$\psi(z,t) = \begin{array}{l}\text{instantaneous transverse displacement of} \\ \quad\text{particles having equilibrium position } z.\end{array} \tag{4}$$

Now consider a very small segment of the continuous string. At equilibrium, the segment occupies a small interval of length Δz centered at z. The mass ΔM of the segment divided by the length Δz is defined as the *mass density* ρ_0, measured in units of mass per unit length:

$$\Delta M = \rho_0\, \Delta z. \tag{5}$$

The mass density is assumed to be uniform along the string. The string tension at equilibrium, denoted by T_0, is also assumed to be uniform.

For a general (nonequilibrium) situation, the segment has a transverse displacement $\psi(z,t)$, averaged over the segment. (See Fig. 2.2.) The segment is no longer exactly straight; it has (generally) a slight curvature. This is indicated in Fig. 2.2 by the fact that θ_1 and θ_2 are not equal. The tension in the segment is no longer T_0, since the segment is longer than its equilibrium length Δz. Let us find the net force F_x on the segment at the instant shown. At its left end, the segment is pulled downward with a force $T_1 \sin \theta_1$. At its right end, it is pulled upward with a force $T_2 \sin \theta_2$. Thus the net force upward is

$$F_x(t) = T_2 \sin \theta_2 - T_1 \sin \theta_1. \tag{6}$$

We want to express $F_x(t)$ in terms of $\psi(z,t)$ and its space derivative

$$\frac{\partial \psi(z,t)}{\partial z} = \text{slope of string at position } z \text{ at time } t. \tag{7}$$

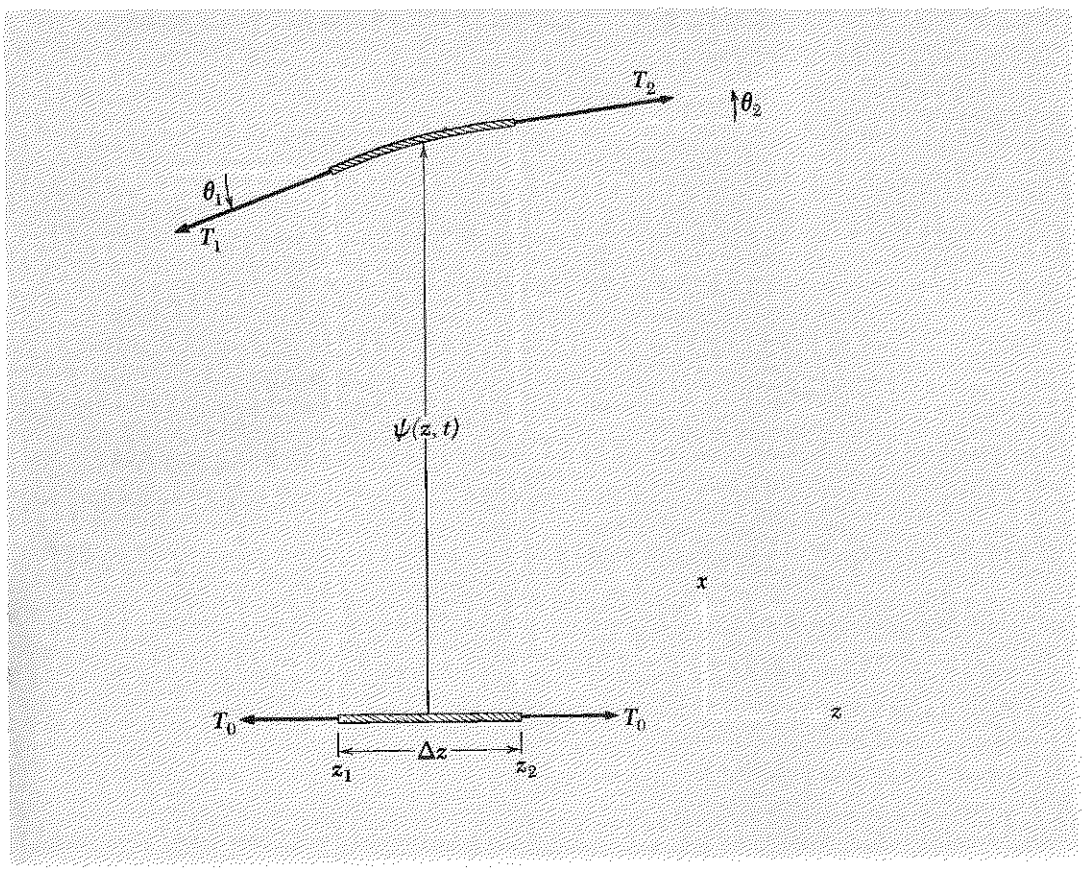


Fig. 2.2 ***Transverse oscillations of a continuous string. At bottom is the equilibrium position of an infinitesimal segment along the z axis. Above is a general position and configuration of the same segment.***

According to Fig. 2.2, the string slope at z_1 is $\tan\theta_1$, and the slope at z_2 is $\tan\theta_2$. Also, $T_1\cos\theta_1$ is the horizontal component of the string tension at z_1, and $T_2\cos\theta_2$ is the horizontal component at z_2. Now, we want eventually to obtain a *linear* differential equation of motion. To this end, we shall assume that we can use either the slinky approximation or the small-oscillations approximation. In the slinky approximation, T is larger than T_0 by a factor $1/\cos\theta$, because the segment is longer than Δz by a factor $1/\cos\theta$. Therefore $T\cos\theta = T_0$. In the small-oscillations approximation, we neglect the increase in length of the segment, and we also approximate $\cos\theta$ by 1. Thus we have $T\cos\theta = T_0$ in that case also. Then Eq. (6) gives

$$\begin{aligned} F_x(t) &= T_2\sin\theta_2 - T_1\sin\theta_1 \\ &= T_2\cos\theta_2\tan\theta_2 - T_1\cos\theta_1\tan\theta_1 \\ &= T_0\tan\theta_2 - T_0\tan\theta_1 \\ &= T_0\left(\frac{\partial\psi}{\partial z}\right)_2 - T_0\left(\frac{\partial\psi}{\partial z}\right)_1. \end{aligned} \tag{8}$$

Now consider the function $f(z)$ defined by

$$f(z) = \frac{\partial\psi(z,t)}{\partial z}, \tag{9}$$

where we have suppressed the variable t in writing $f(z)$ because we intend to hold t constant. We expand $f(z)$ in a Taylor's series around z_1 and then set $z = z_2$. [See Appendix Eq. (3)]:

$$f(z_2) = f(z_1) + (z_2 - z_1)\left(\frac{df}{dz}\right)_1 + \frac{1}{2}(z_2 - z_1)^2\left(\frac{d^2f}{dz^2}\right)_1 + \cdots, \tag{10}$$

where $z_2 - z_1 = \Delta z$, according to Fig. 2.2. We now go to the limit in which Δz is small enough so that we can neglect quadratic and higher terms in Eq. (10). Then we write

$$\begin{aligned} f(z_2) - f(z_1) = \Delta z\left(\frac{df}{dz}\right)_1 &= \Delta z\frac{d}{dz}\left(\frac{\partial\psi(z,t)}{\partial z}\right) \\ &= \Delta z\frac{\partial}{\partial z}\left(\frac{\partial\psi(z,t)}{\partial z}\right) \\ &= \Delta z\frac{\partial^2\psi(z,t)}{\partial z^2}. \end{aligned} \tag{11}$$

Notice that in arriving at Eq. (11) we dropped the subscript 1. That is because it does not matter where in the interval Δz we evaluate the z derivative, since we are neglecting higher derivatives in the Taylor's series, Eq. (10). Notice also that we must write the space derivative as a partial derivative once we use the notation $\psi(z,t)$.

We can now use Eqs. (9) and (11) in Eq. (8) to obtain for the net force on the segment the result

$$F_x(t) = T_0\,\Delta_z \frac{\partial^2\psi(z,t)}{\partial z^2}. \tag{12}$$

We now use Newton's second law. The force F_x, as given by Eq. (12), equals the mass ΔM of the segment times the acceleration of the segment. The velocity and acceleration of the segment with equilibrium position z are expressed in terms of $\psi(z,t)$ and its derivatives as follows:

$$\begin{aligned} \psi(z,t) &= \text{displacement} \\ \frac{\partial\psi(z,t)}{\partial t} &= \text{velocity} \\ \frac{\partial^2\psi(z,t)}{\partial t^2} &= \text{acceleration.} \end{aligned} \tag{13}$$

Thus Newton's law [with $\Delta M = \rho_0\,\Delta z$] gives

$$\rho_0\,\Delta z \frac{\partial^2\psi}{\partial t^2} = F_x = T_0\,\Delta z \frac{\partial^2\psi}{\partial z^2},$$

i.e.,

$$\boxed{\frac{\partial^2\psi(z,t)}{\partial t^2} = \frac{T_0}{\rho_0}\frac{\partial^2\psi(z,t)}{\partial z^2}} \tag{14}$$

Classical wave equation. Equation (14) is a very famous second-order linear partial differential equation. It is called the *classical wave equation.* We will encounter it often and will eventually know many of the properties of its solutions and the physical situations where it occurs. (Of course the positive constant T_0/ρ_0 is special to the string. In other physical applications, some other positive constant appears in its place in the wave equation.)

Standing waves. We are trying to find the normal modes—the standing waves—of a continuous string. Therefore we *assume* that we have a mode. We assume that all parts of the string oscillate in harmonic motion at the same angular frequency ω and with the same phase constant φ. Thus $\psi(z,t)$, which is the displacement of string particles with equilibrium position z, should have the same time dependence, $\cos(\omega t + \varphi)$ for all particles, i.e., for all z. As usual, the phase constant φ corresponds to the "turn-on time" of the mode. The "shape" of a mode made up of discrete degrees of freedom labeled a, b, c, etc., is given by the relative vibration amplitudes, A, B, C, etc. In the present case of a continuous string, where the (infinitely many) degrees of freedom are labeled by the parameter z,

the amplitude of vibration of the degrees of freedom at z (i.e., in a small neighborhood of z) can be written as a continuous function of z denoted by $A(z)$. The shape of $A(z)$ as a function of z depends on the mode; that is, each mode has a different $A(z)$. Thus we can write down the *general expression for a standing wave:*

$$\psi(z,t) = A(z)\cos(\omega t + \varphi). \tag{15}$$

The acceleration corresponding to Eq. (15) is

$$\frac{\partial^2\psi}{\partial t^2} = -\omega^2\psi = -\omega^2 A(z)\cos(\omega t + \varphi) \tag{16}$$

The second partial derivative of Eq. (15) with respect to z is

$$\begin{aligned}\frac{\partial^2\psi}{\partial z^2} &= \frac{\partial^2[A(z)\cos(\omega t + \varphi)]}{\partial z^2}\\ &= \cos(\omega t + \varphi)\frac{d^2A(z)}{dz^2},\end{aligned} \tag{17}$$

where we have an ordinary derivative with respect to z rather than a partial derivative because $A(z)$ has no time dependence. Inserting Eqs. (16) and (17) into Eq. (14) and canceling the common factor $\cos(\omega t + \varphi)$, we obtain

$$\frac{d^2A(z)}{dz^2} = -\omega^2\frac{\rho_0}{T_0}A(z). \tag{18}$$

Equation (18) governs the shape of the mode. Since each mode has a different angular frequency ω, and since ω^2 appears in Eq. (18), we see that different modes have different shapes, as expected.

Equation (18) is of the form of the differential equation for harmonic oscillation, but for oscillation in space rather than in time. The general form of a harmonic oscillation in space can be written

$$A(z) = A\sin\left(2\pi\frac{z}{\lambda}\right) + B\cos\left(2\pi\frac{z}{\lambda}\right), \tag{19}$$

where the constant λ represents the distance over which one complete oscillation occurs. Thus it is called the *wavelength*. It is the parameter for oscillations in space analogous to the period T for oscillations in time. The wavelength λ is measured in units of centimeters per cycle (i.e., per cycle of spatial oscillation along z), or simply in centimeters.

To see how to adapt this solution to Eq. (18), differentiate Eq. (19) twice:

$$\frac{d^2A(z)}{dz^2} = -\left(\frac{2\pi}{\lambda}\right)^2 A(z). \tag{20}$$

Then comparing Eqs. (18) and (20), we see that we need to have

$$\left(\frac{2\pi}{\lambda}\right)^2 = \omega^2\left(\frac{\rho_0}{T_0}\right) = (2\pi\nu)^2\frac{\rho_0}{T_0}, \tag{21}$$

i.e.,

$$\boxed{\lambda\nu = \sqrt{\frac{T_0}{\rho_0}} \equiv v_0 = \text{constant.}} \tag{22}$$

Wave velocity. Equation (22) gives the relation between wavelength and frequency for transverse standing waves on a continuous homogeneous string. The constant $(T_0/\rho_0)^{1/2}$ has the dimensions of velocity, since $\lambda\nu$ has dimensions length/time. The velocity $v_0 \equiv (T_0/\rho_0)^{1/2}$ is called the "phase velocity for traveling waves," for this system. (We will study traveling waves in Chap. 4.) In our present study of standing waves, the concept of phase velocity is not needed, because standing waves do not "go anywhere." They "stand and wave" like a big "distributed" harmonic oscillator. Hereafter in this chapter we shall avoid calling $(T_0/\rho_0)^{1/2}$ a velocity, because we want your mental picture to be that of standing waves.

The general solution for the displacement $\psi(z,t)$ of the string in a single mode (standing wave) is obtained by combining Eqs. (15) and (19):

$$\psi(z,t) = \cos(\omega t + \varphi)[A\sin(2\pi z/\lambda) + B\cos(2\pi z/\lambda)]. \tag{23}$$

Boundary conditions. Equation (23) is slightly *too* general. It does not manifest the important boundary conditions. Our vibrating string is *fixed at both ends,* but we have not yet incorporated that bit of information into the solution. We do so as follows. Suppose the string has total length L. Let us choose the origin of coordinates so that the left-hand end of the string is at $z = 0$. The right-hand end is then at $z = L$. Consider $z = 0$. The string is fixed there, so $\psi(0,t)$ must be zero for all t. This condition requires that $B = 0$, since, for all times t,

$$\psi(0,t) = \cos(\omega t + \varphi)[0 + B] = 0. \tag{24}$$

Thus we have

$$\psi(z,t) = A\cos(\omega t + \varphi)\sin\frac{2\pi z}{\lambda}. \tag{25}$$

The other boundary condition is that the string be fixed at $z = L$, so $\psi(L,t)$ must be zero for all t. We certainly do not want to choose $A = 0$ in Eq. (25), since that corresponds to the uninteresting situation of a string

permanently at rest. The only way we can satisfy the boundary condition at L is to have

$$\sin \frac{2\pi L}{\lambda} = 0. \tag{26}$$

The only wavelengths λ that can satisfy this boundary condition are those for which the number of half-wavelengths, L, is an integer. Thus the acceptable wavelengths must satisfy one of the following possibilities:

$$\frac{2\pi L}{\lambda} = \pi, 2\pi, 3\pi, 4\pi, 5\pi, \ldots. \tag{27}$$

(Why did we exclude the case $2\pi L/\lambda = 0$?) This sequence of possible ways to satisfy the boundary conditions corresponds to all the possible modes of the string. We number the modes according to the sequence, beginning with the first term in the sequence as number 1. Then according to Eq. (27), we have the wavelengths of the modes given by

$$\lambda_1 = 2L, \qquad \lambda_2 = \tfrac{1}{2}\lambda_1, \qquad \lambda_3 = \tfrac{1}{3}\lambda_1, \qquad \lambda_4 = \tfrac{1}{4}\lambda_1, \qquad \ldots. \tag{28}$$

Harmonic frequency ratios. The corresponding frequencies of the modes are found by using Eq. (22):

$$\nu_1 = \frac{v_0}{\lambda_1}, \qquad \nu_2 = 2\nu_1, \qquad \nu_3 = 3\nu_1, \qquad \nu_4 = 4\nu_1, \qquad \ldots. \tag{29}$$

The frequencies $2\nu_1$, $3\nu_1$, etc., are called the second, third, etc., *harmonics* of the *fundamental* frequency ν_1. The fact that the mode frequencies ν_2, ν_3, etc., consist of a sequence of *harmonics* of the lowest mode frequency ν_1 is a result of our assumption that the string is perfectly uniform and flexible. Most real physical systems have mode frequencies that do not follow this harmonic sequence of frequency ratios. For example, the mode frequencies for a string of nonuniform mass density do not form a sequence of harmonics of the fundamental. Instead one might have, for example, $\nu_2 = 2.78\nu_1$, $\nu_3 = 4.62\nu_1$, etc. For a real piano or violin string, the mode requencies follow approximately, but not exactly, the harmonic sequence. That is because they are not perfectly flexible. (For a qualitative argument that shows how these "harmonic" frequency ratios are due to the uniformity of the string, see Prob. 2.7.)

The modes of the string are shown in Fig. 2.3. The equilibrium configuration would correspond to the missing first term, $2\pi L/\lambda = 0$, in the sequence given by Eq. (27). The corresponding frequency is zero. There is no motion, and the equilibrium state is not called a mode.

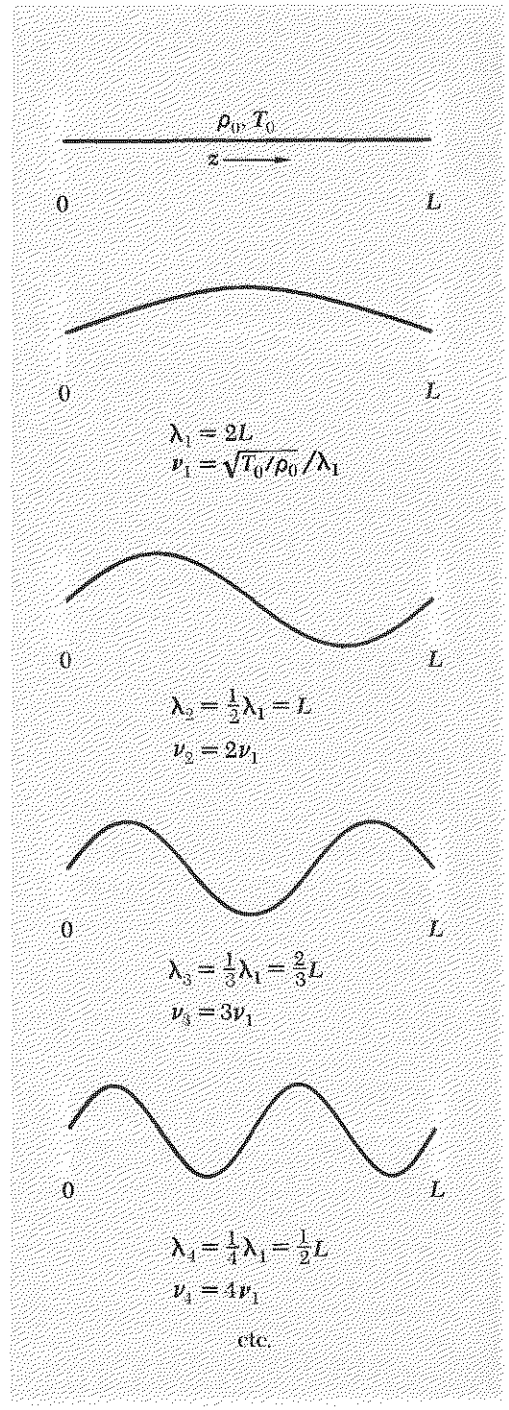


Fig. 2.3 ***Modes of continuous homogeneous string with fixed ends.***

Wavenumber. The inverse of the wavelength λ is called the *wavenumber* σ. Its units are cycles per centimeter or, more often, "inverse centimeters." It is the parameter for oscillations in space analogous to the frequency ν for oscillations in time.

$$\sigma = \frac{1}{\lambda} = \text{wavenumber (cycles per cm)}. \tag{30}$$

The wavenumber times 2π is called the *angular wavenumber* k. Its units are radians of phase per centimeter. It is the quantity for oscillations in space analogous to the angular frequency ω for oscillations in time.

$$k = \frac{2\pi}{\lambda} = \text{angular wavenumber (radians per cm)}. \tag{31}$$

We can illustrate the use of these quantities by writing the same standing wave in several equivalent forms:

$$\psi(z,t) = A \sin 2\pi \frac{t}{T} \sin 2\pi \frac{z}{\lambda} = A \sin 2\pi\nu t \sin 2\pi\sigma z = A \sin \omega t \sin kz. \tag{32}$$

As another illustration, we can describe the sequence of normal modes given by Eqs. (27), (28), and (29) as follows:

$$k_1 L = \pi \text{ rad}, \qquad k_2 L = 2\pi \text{ rad}, \qquad k_3 L = 3\pi \text{ rad}, \qquad \text{etc.} \tag{33}$$

$$\sigma_1 L = \tfrac{1}{2} \text{ cycle}, \qquad \sigma_2 L = 1 \text{ cycle}, \qquad \sigma_3 L = \tfrac{3}{2} \text{ cycle}, \qquad \text{etc.} \tag{34}$$

Dispersion relation. Equation (22) gives the relation between frequency and wavelength for the normal modes of the uniform flexible string:

$$\nu = \sqrt{\frac{T_0}{\rho_0}} \cdot \frac{1}{\lambda} = \sqrt{\frac{T_0}{\rho_0}} \cdot \sigma,$$

or (multiplying by 2π)

$$\omega = \sqrt{\frac{T_0}{\rho_0}}\, k. \tag{35}$$

Equation (35) gives the relation between frequency and wavenumber for the normal modes of the string. (Note that we dropped the adjective "angular" from the designations "angular frequency" and "angular wavenumber." This is common practice, but the symbols and the units always remove any ambiguity.) Such a relation, giving ω as a function of k, is called a *dispersion relation*. It is a convenient way of characterizing the wave behavior of a system.

Dispersion law for real piano string. The dispersion relation given by Eq. (35) is extremely simple, but we shall find more complicated ones later. For a more complicated dispersion relation, the quantity $\lambda\nu = \omega/k$ is *not* constant, i.e., it is not independent of wavelength. For example, it turns out that the dispersion law for a real piano string is given approximately by

$$\frac{\omega^2}{k^2} = \frac{T_0}{\rho_0} + \alpha k^2 \tag{36}$$

where α is a small positive constant that would be zero if the string were perfectly flexible. [In that case Eq. (36) reduces to Eq. (35).] The modes of a real piano string have the same spatial dependence as those of a perfectly flexible string, i.e., $\lambda_1 = 2L$, $\lambda_2 = \frac{1}{2}\lambda_1$, $\lambda_3 = \frac{1}{3}\lambda_1$, etc., because the boundary conditions are the same. But the mode frequencies do *not* satisfy the "harmonic" sequence $\nu_2 = 2\nu_1$, $\nu_3 = 3\nu_1$, etc., because the dispersion relation Eq. (36) does not give that sequence. The harmonic sequence is obtained only in the idealized limit where α is zero, i.e., where we have $\lambda\nu =$ constant. For a real piano string the frequencies of the higher modes are slightly "sharper" (i.e., have slightly higher frequencies) than the frequencies given by the harmonic sequence.

Nondispersive and dispersive waves. Waves satisfying the simple dispersion relation $\omega/k =$ constant are called "nondispersive waves." When ω/k depends on the wavelength (and hence on the frequency), the waves are called "dispersive." For dispersive waves, it is customary to make a plot of ω versus k. In the present example of the flexible string this plot is just a straight line passing through the point $\omega = k = 0$ and having slope $(T_0/\rho_0)^{1/2}$, as shown in Fig. 2.4.

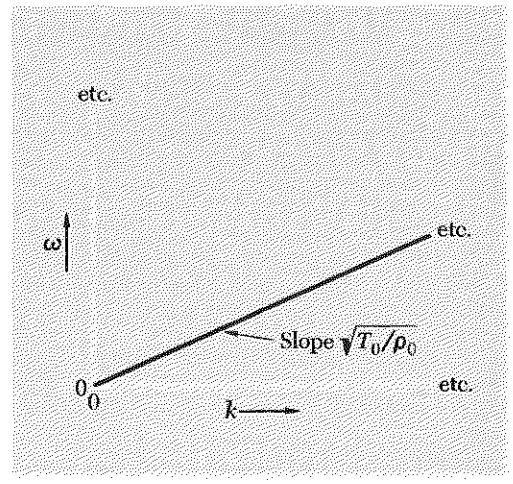


Fig. 2.4 Dispersion relation for continuous, homogeneous, flexible string.

2.3 *General Motion of Continuous String and Fourier Analysis*

The most general state of motion of the continuous string (with both ends fixed and for transverse vibrations along x) is given by a superposition of all the modes, numbered 1, 2, 3, . . . , with amplitudes $A_1, A_2, A_3, \ldots$, and phase constants $\varphi_1, \varphi_2, \varphi_3, \ldots$:

$$\psi(z,t) = A_1 \sin k_1 z \cos(\omega_1 t + \varphi_1) + A_2 \sin k_2 z \cos(\omega_2 t + \varphi_2) + \cdots, \tag{37}$$

where k_n are chosen as described in the preceding section to satisfy the boundary conditions at $z = 0$ and $z = L$, and where ω_n are related to k_n by the dispersion relation $\omega(k)$. The amplitudes A_n and phase constants φ_n, which complete the description of the motion for all positions z and times t, are determined by specifying the *initial conditions*, namely, the instantaneous displacement $\psi(z,t)$ and the corresponding instantaneous velocity $v(z,t) = \partial\psi(z,t)/\partial t$ for each point z at $t = 0$.

Motion of string fixed at both ends. Suppose that for $t < 0$ we constrain the string to follow a prescribed shape $f(z)$ by means of some sort of template. Then, at $t = 0$, we let the string go by suddenly removing the template. Thus at $t = 0$ each part of the string has its displacement $\psi(z,0)$ equal to $f(z)$ and has velocity $v(z,0)$ equal to zero. Now, the nth term in the velocity [which is the time derivative of Eq. (37)] is proportional to $\sin(\omega_n t + \varphi_n)$, which reduces to $\sin \varphi_n$ at $t = 0$. Thus we can make $v(z,0) = 0$ for all z simply by setting each phase constant φ_n equal either to zero or to π. However, the phase constant $\varphi_1 = \pi$ (for example) is just equivalent to a minus sign affixed to A_1. Therefore we can satisfy these initial conditions if we set all the phase constants to zero but allow the amplitudes A_1, A_2, etc., to be either positive or negative. Thus we have, for $v(z,0) = 0$,

$$\psi(z,t) = A_1 \sin k_1 z \cos \omega_1 t + A_2 \sin k_2 z \cos \omega_2 t + \cdots, \tag{38}$$

and, at $t = 0$,

$$\psi(z,0) = f(z) = A_1 \sin k_1 z + A_2 \sin k_2 z + \cdots. \tag{39}$$

As we shall see below, Eq. (39) determines the amplitudes $A_1, A_2, \ldots$.

Fourier series for function with zeros at both ends. Now, the function $f(z)$ can be a very general function of z. The only condition we specified was that it was to constrain the string. Therefore, virtually all we require of $f(z)$ is that we have $f(z) = 0$ at $z = 0$ and $z = L$. We also require that $f(z)$ not be "jagged" on a "small" scale, since our wave function $\psi(z,t)$ is supposed to be a slowly varying function of z. Therefore, $f(z)$ must be reasonably smooth in order for us to be able to use it to constrain the string and still have the string obey the differential equation that we obtained in the "continuous" approximation. Thus we have found that *any* reasonable function $f(z)$ that vanishes at $z = 0$ and L can be expanded in a series of the form of Eq. (39), i.e., as a sum of sinusoidal oscillations. Equation (39) is called a *Fourier series* or *Fourier expansion.* It is a special example of a Fourier series in that it applies only to functions $f(z)$ that vanish at $z = 0$ and L. However, a much broader class of functions can be expressed in appropriate Fourier expansions. We shall now find this broader class of functions.

Our function $f(z)$ was used to constrain the string, and therefore it was defined only between $z = 0$ and L. However, the functions $\sin k_1 z$, $\sin 2k_1 z$, $\sin 3k_1 z$, etc., that make up the infinite series of Eq. (39) are defined for all z from $-\infty$ to $+\infty$. Also, we notice that $\sin k_1 z$ is *periodic* in z with period λ_1. This means it satisfies the *periodicity condition*, namely, that for any given z, it must have the same value at $z + \lambda_1$ as it does at z. (The period λ_1 is $2L$ in our example.) We notice that the function $\sin 2k_1 z$ is also periodic in z with period λ_1. (Of course it goes through two cycles in distance λ_1; it is thus periodic with period $\frac{1}{2}\lambda_1$, as well as periodic with

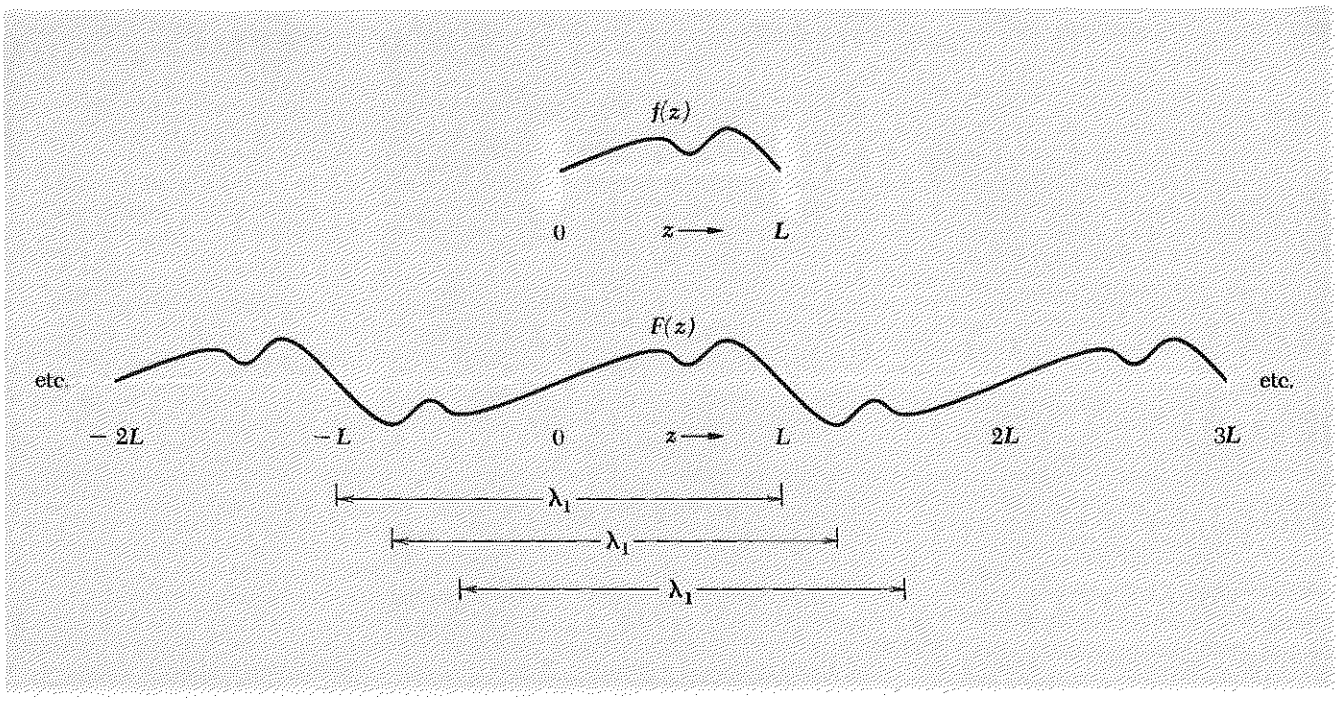


Fig. 2.5 Construction of a periodic function F(z) with period $\lambda_1 = 2L$ from a function f(z) that vanishes at z = 0 and L. Note that F(z) satisfies the periodicity condition.

period λ_1.) In fact, all the sinusoidal functions in the expansion, Eq. (39), are periodic in z with period λ_1. Therefore, the expansion itself is periodic with period λ_1. Thus we can broaden the class of functions which have a Fourier expansion of the form of Eq. (39): all periodic functions $F(z)$ with period λ_1 that vanish at $z = 0$ and at $z = \frac{1}{2}\lambda_1$ can be expanded in a Fourier series of the form of Eq. (39). Given a function $f(z)$ defined only between $z = 0$ and L and vanishing at those points, we can construct a periodic function $F(z)$ which will have the same Fourier expansion as $f(z)$ by the following procedure: Between $z = 0$ and L, we let $F(z)$ coincide with $f(z)$. Between L and $2L$, we construct $F(z)$ by making an "inverted mirror image" of $f(z)$ in a "mirror" located at $z = L$. Now that we have defined $F(z)$ between $z = 0$ and $2L$, we simply repeat it in successive intervals of length $2L$ to define $F(z)$ for all z. The construction is shown in Fig. 2.5.

Fourier analysis of a periodic function of z. We now broaden the class of functions for which we can write Fourier expansions once more, as follows: Equation (39) corresponds only to functions that are periodic with period λ_1 and that vanish at $z = 0$ and $\frac{1}{2}\lambda_1$. However, the condition that the function vanish at $z = 0$ and $\frac{1}{2}\lambda_1$ was the result of our particular choice of boundary conditions, namely that the string have both ends fixed. Without those particular boundary conditions, we would have obtained solutions for the string vibrations which included not only the terms in $\sin mk_1z$ but also terms in $\cos mk_1z$. These functions are also periodic in z with period λ_1,

but they do not vanish at $z = 0$ and $\frac{1}{2}\lambda_1$. (They correspond to string vibrations with a free end or ends.) By including them in the series, we finally arrive at a very general class of functions for which we can write Fourier series: all (reasonable) periodic functions $F(z)$ with period λ_1, i.e., functions such that $F(z + \lambda_1) = F(z)$ for all z, can be expanded in a Fourier series of the form

$$\begin{aligned} F(z) &= \sum_{n=0}^{\infty} \left[A_n \sin n\frac{2\pi}{\lambda_1} z + B_n \cos n\frac{2\pi}{\lambda_1} z \right] \\ &= B_0 + \sum_{n=1}^{\infty} A_n \sin n\frac{2\pi}{\lambda_1} z + \sum_{n=1}^{\infty} B_n \cos n\frac{2\pi}{\lambda_1} z \\ &= B_0 + \sum_{n=1}^{\infty} A_n \sin nk_1 z + \sum_{n=1}^{\infty} B_n \cos nk_1 z. \end{aligned} \tag{40}$$

Finding Fourier coefficients. The process of finding the amplitudes or *Fourier coefficients* B_0, A_n, and B_n (for all n) for a given periodic function $F(z)$ is called *Fourier analysis.* We shall now show you how to find these coefficients.

First we find B_0 as follows: We integrate both sides of Eq. (40) over any complete period of $F(z)$; i.e., we integrate from $z = z_1$ to $z = z_2$, where z_1 is any value of z and where $z_2 = z_1 + \lambda_1$. The function $F(z)$ is assumed to be known; therefore its integral from z_1 to z_2, which is the integral of the left side of Eq. (40), can be found. Now consider the integral of the right side of Eq. (40). There are an infinite number of terms and therefore an infinite number of integrals to consider. The first term is B_0; it produces the integral

$$\int_{z_1}^{z_2} B_0\, dz = B_0(z_2 - z_1) = B_0\lambda_1. \tag{41}$$

All the other terms give zero when integrated over one period. That is because $\sin nk_1z$ and $\cos nk_1z$ are as often negative as positive in any complete period, and therefore they integrate to zero:

$$\int_{z_1}^{z_2} \sin nk_1z\, dz = 0; \qquad \int_{z_1}^{z_2} \cos nk_1z\, dz = 0.$$

Thus we have found B_0. It is given by

$$B_0\lambda_1 = \int_{z_1}^{z_2} F(z)\, dz. \tag{42}$$

Next we show you how to find A_m, where m is any particular value of n in Eq. (40) from 1 to infinity. The trick is to multiply both sides of Eq. (40) by $\sin mk_1z$ and integrate both sides over one complete period of $F(z)$. The integral of the left-hand side can be evaluated since $F(z)$ is known. Now consider the integral of the right-hand side. The first term is the integral of B_0 times $\sin mk_1z$; that integrates to zero because it includes m complete

periods of $\sin mk_1z$. That leaves us with the integrals of $\sin nk_1z \sin mk_1z$ and of $\cos nk_1z \sin mk_1z$ for $n = 1, 2, \ldots$. Consider the particular term that has $n = m$. The square of $\sin mk_1z$ averages to $\frac{1}{2}$ over one period of $F(z)$ of length λ_1 (which is m complete periods of the function $\sin mk_1z$). This gives a contribution $\frac{1}{2}A_m\lambda_1$ to the integral of the right side of Eq. (40). All other terms contribute zero. We see that as follows: Consider for example the integrand $\sin nk_1z \sin mk_1z$, for m not equal to n. This can be written in the form

$$\sin nk_1z \sin mk_1z = \tfrac{1}{2}\cos(n - m)k_1z - \tfrac{1}{2}\cos(n + m)k_1z. \tag{43}$$

Since $n - m$ and $n + m$ are integers, each of the two terms on the right side of Eq. (43) is as often positive as negative in any complete period of $F(z)$ of length λ_1. Therefore both terms integrate to zero (except for the case $n = m$ which we have already considered). Similarly, the terms of the form $\cos nk_1z \sin mk_1z$ integrate to zero because of the identity

$$\cos nk_1z \sin mk_1z = \tfrac{1}{2}\sin(m + n)k_1z + \tfrac{1}{2}\sin(m - n)k_1z.$$

Thus we find that

$$\frac{1}{2}A_m\lambda_1 = \int_{z_1}^{z_2} \sin mk_1z\, F(z)\, dz. \tag{44}$$

Similarly, we can find the coefficients B_m by multiplying both sides of Eq. (40) by $\cos mk_1z$ and integrating over one period of length λ_1. The only nonzero contribution to the integral of the right side comes from the term with coefficient B_m. Thus we find that

$$\frac{1}{2}B_m\lambda_1 = \int_{z_1}^{z_2} \cos mk_1z\, F(z)\, dz. \tag{45}$$

Fourier coefficients. Our results are given by Eqs. (40), (42), (44), and (45), which we collect in one place for convenience of future reference:

$$\boxed{\begin{aligned} F(z) &= B_0 + \sum_{m=1}^{\infty} A_m \sin mk_1z + \sum_{m=1}^{\infty} B_m \cos mk_1z, \\ B_0 &= \frac{1}{\lambda_1}\int_{z_1}^{z_1+\lambda_1} F(z)\, dz, \\ A_m &= \frac{2}{\lambda_1}\int_{z_1}^{z_1+\lambda_1} F(z) \sin mk_1z\, dz, \\ B_m &= \frac{2}{\lambda_1}\int_{z_1}^{z_1+\lambda_1} F(z) \cos mk_1z\, dz, \end{aligned}} \tag{46}$$

where z_1 is any value of z. Equations (46) tell us how to Fourier-analyze $F(z)$, any periodic function of z having period λ_1.

Square wave. Here is an illustrative example, the Fourier analysis of a "square wave." Let $f(z)$ be zero at the points $z = 0$ and $z = L$, but let it equal $+1$ for $0 < z < L$. (This function has a discontinuity at $z = 0$ and another at $z = L$, so that it does not satisfy the assumption in our discussion above that it be "smooth" everywhere. Therefore we cannot reasonably expect the Fourier series to give a perfect representation of a square wave. It turns out that there is a sharp "overshoot spike" at $z = 0$ and at $z = L$ for every partial sum of the series. As more and more terms are added, the spike gets sharper, but its height does not go to zero.)

The periodic function $F(z)$ that we construct according to the prescription of Fig. 2.5 is given as follows: $F(z) = 0$ for $z = 0$; $+1$ for $0 < z < L$; 0 for $z = L$; -1 for $L < z < 2L$; etc.; as shown in Fig. 2.6.

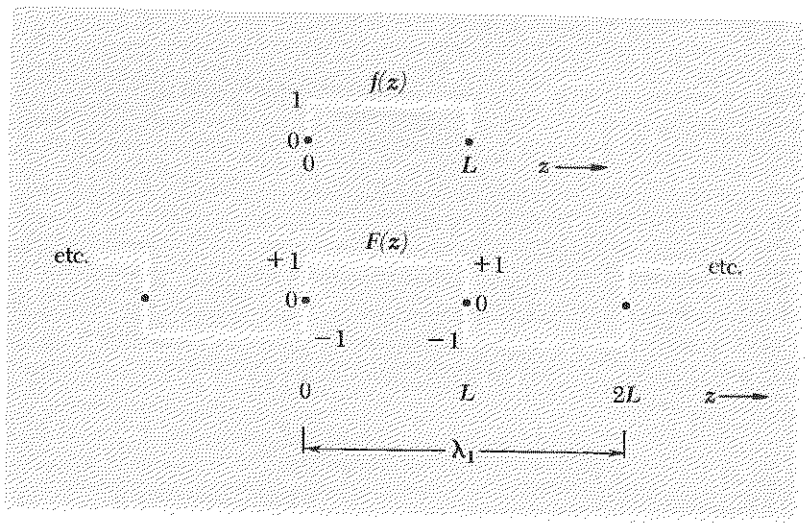


Fig. 2.6 ***Square wave $f(z)$. Periodic square wave $F(z)$.***

Using Eqs. (46), one can easily obtain the results (Prob. 2.11) $B_0 = 0$; $B_m = 0$ for all m; $A_m = 0$ for $m = 2, 4, 6, 8 \ldots$ (even integers); $A_m = 4/m\pi$ for $m = 1, 3, 5, 7, \ldots$ (odd integers). Thus $F(z)$ is given by

$$F(z) = B_0 + \sum_{m=1}^{\infty} B_m \cos mk_1 z + \sum_{m=1}^{\infty} A_m \sin mk_1 z$$

$$= \frac{4}{\pi}\left\{\sin k_1 z + \frac{1}{3}\sin 3k_1 z + \frac{1}{5}\sin 5k_1 z + \cdots\right\}$$

$$= 1.273 \sin\frac{\pi z}{L} + 0.424 \sin\frac{3\pi z}{L} + 0.255 \sin\frac{5\pi z}{L} + \cdots. \tag{47}$$

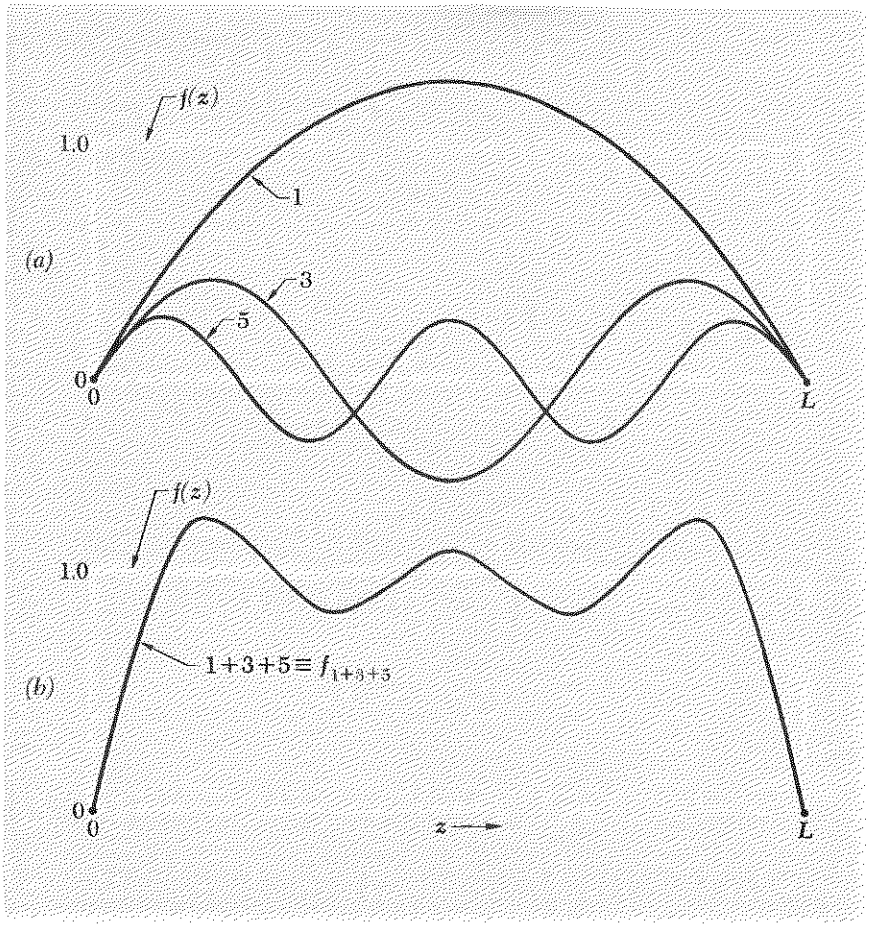


Fig. 2.7 Fourier analysis of square wave f(z). (a) Square wave f(z) and the first three contributions to its Fourier decomposition. The labels 1, 3, and 5 refer to the normal modes 1, 3, 5. (b) Square wave f(z) and the superposition f_{1+3+5} of its first three Fourier components.

In Fig. 2.7 are shown the square wave $f(z)$, the first three contributing terms given by Eq. (47), and the superposition of these first three terms.

Suppose that instead of trying to force a slinky into the configuration of the sharp-cornered function $f(z)$ which we have been considering, we constrain it at time zero to follow exactly the function

$$g(z) = 1.273 \sin \frac{\pi z}{L} + 0.424 \sin \frac{3\pi z}{L} + 0.255 \sin \frac{5\pi z}{L}. \tag{48}$$

This corresponds to the first three terms of Eq. (47) and is plotted in Fig. 2.7*b*. Now we let the slinky go at $t = 0$. What is $\psi(z,t)$? Does the shape remain constant as t increases? (See Prob. 2.16.)

Fourier analysis of a periodic function of time. Suppose we are given a function $F(t)$ that is defined for all t and that is periodic in t with period T_1:

$$F(t + T_1) = F(t) \qquad \text{for any } t. \tag{49}$$

We assume that $F(t)$ can be expanded in the Fourier series

$$F(t) = B_0 + \sum_{n=1}^{\infty} A_n \sin n\omega_1 t + \sum_{n=1}^{\infty} B_n \cos n\omega_1 t, \tag{50}$$

with

$$\omega_1 = 2\pi\nu_1 = \frac{2\pi}{T_1}. \tag{51}$$

The Fourier coefficients can be obtained directly from our results for the Fourier analysis of a spatially periodic function $F(z)$, which we studied above. The mathematical analysis cannot distinguish the variable $\theta = \omega_1 t$ from the variable $\theta = k_1 z$. Thus we obtain the results for the coefficients in Eqs. (50) directly from Eqs. (46):

$$\begin{aligned} B_0 &= \frac{1}{T_1}\int_{t_1}^{t_1+T_1} F(t)\, dt, \\ B_n &= \frac{2}{T_1}\int_{t_1}^{t_1+T_1} F(t) \cos n\omega_1 t\, dt, \\ A_n &= \frac{2}{T_1}\int_{t_1}^{t_1+T_1} F(t) \sin n\omega_1 t\, dt, \end{aligned} \tag{52}$$

where the time t_1 is any convenient time.

Sound of a piano chord. We shall illustrate this with a superposition of known ingredients, rather than by a Fourier analysis of a known function $F(t)$. Suppose you have a piano that is tuned to the "scientific scale." (See Home Exp. 2.6 if you want to know more about musical scales.) Let $\nu_1 = 128$ cps. That is the note C one octave (i.e., a factor of 2 in frequency) below middle C. Now let $\nu_3 = 3\nu_1 = 384$ cps. That is the G above middle C. Let $\nu_5 = 5\nu_1 = 640$ cps. That is the E above the G above middle C. Now strike all three notes at the same time. One hears a nice "open" chord. If you strike them at exactly the same time, and if you adjust your striking force so that the gauge pressure of air produced at your ear by the C128 string is (in appropriate units) $1.273 \sin 2\pi\nu_1 t$, pressure by the G384 string is $0.424 \sin 2\pi\nu_3 t$, and pressure by the E640 string is $0.255 \sin 2\pi\nu_5 t$, then the total air pressure $p(t)$ at your ear is the superposition

$$p(t) = 1.273 \sin 2\pi\nu_1 t + 0.424 \sin 2\pi\nu_3 t + 0.255 \sin 2\pi\nu_5 t. \tag{53}$$

But Eq. (53) is very similar to Eq. (48), which is plotted in Fig. 2.7*b*. All we have to do to obtain a plot of $p(t)$ is to change variables from $k_1 z$ to $\omega_1 t$

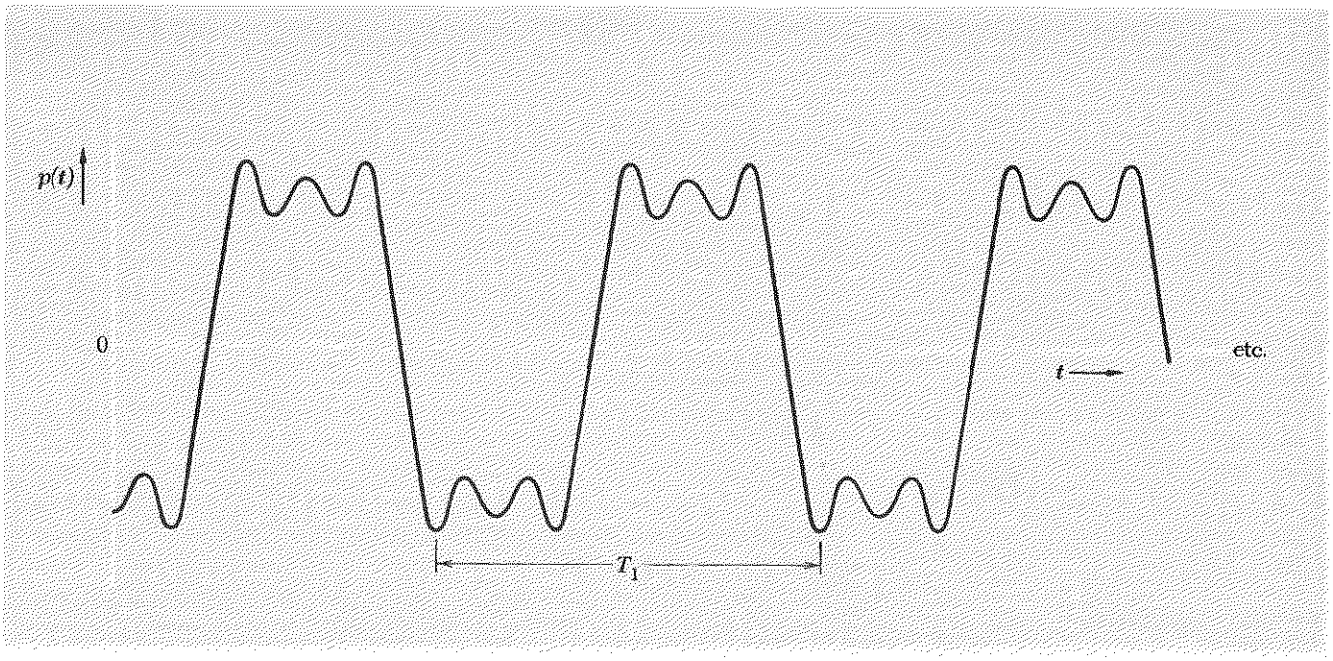


Fig. 2.8 Gauge pressure at ear due to superposition of the notes C128, G384, and E640 with the relative amplitudes and phases of Eq. (53). The period T_1 is (1/128) sec.

and extend the plot shown in Fig. 2.7*b*. Thus we get the result shown in Fig. 2.8.

If we do not strike all the keys at exactly the same time (i.e., to within an accuracy of much less than $\frac{1}{128}$ sec), the relative phases of the three notes will not be those of Eq. (53), and the superposition will not look like Fig. 2.8. But your ear does not notice this! Your ear (plus brain) performs a Fourier analysis on the total pressure. That must be so, because you "hear" the individual notes of the chord and recognize them. But the information as to relative phase of the notes is apparently discarded or perhaps not obtained. Otherwise you would notice a difference in the sound depending on the relative phases.

The pitch-detecting device in the ear is called the *basilar membrane.* It is enclosed in a fluid-filled, spiral-shaped organ in the inner ear called the *cochlea.* The cochlea is mechanically coupled to the eardrum. The end of the basilar membrane nearest the eardrum resonates at about 20,000 cps; the end farthest from the drum resonates at about 20 cps. Thus the extreme range of audible frequencies is about 20 cps to 20 kc. The cochlear nerve has sensors in the basilar membrane and "transduces" the mechanical vibrations into electrical signals that are carried to the brain, where they are somehow processed to become our hearing sensations. By doing the experiment of hitting the chord over and over and seeing that our sensation is the

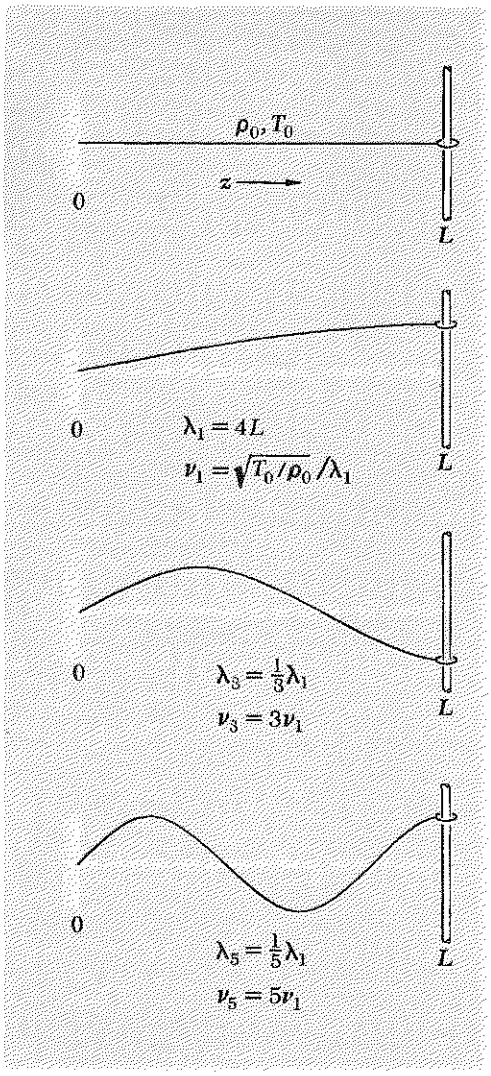


Fig. 2.9 ***Modes of continuous string with one fixed end and one free end.***

same [even though $p(t)$ must have a very different shape depending on the relative phases], we have learned that somewhere the information as to the relative phase of the vibrations of different parts of the basilar membrane is lost. Perhaps this information is never picked up. Perhaps the transducer is a *square-law detector*, i.e., one that puts out an electrical signal proportional to the square of the amplitude of vibration of the membrane. Or perhaps the nerve signal does carry phase information [i.e., perhaps the signal does give $\psi(z,t)$ rather than $\psi^2(z,t)$], but the brain does not use the phase information, i.e., it does not form a superposition of $\psi(z,t)$ from different nerve signals. Apparently there is not much survival value in the phase information; otherwise in our evolutionary development we surely would have acquired some phase-detecting mechanism.

Other boundary conditions. In the general problem of transverse vibrations of a continuous string, it is not necessary that the string be fixed at both ends. One or both ends can be "free," at least as far as transverse oscillations are concerned. The tension and equilibrium configuration of the string can be maintained by a constraint in the form of a massless, frictionless ring sliding on a fixed rod oriented along x, i.e., transverse to the equilibrium axis of the string (which we always take along z). The normal modes will then have different configurations from those we obtained for the string with both ends fixed. The shapes of the modes are still sinusoidal functions of z, as given by Eq. (19). The dispersion relation between frequency and wavelength is still that given by Eq. (22). In fact our entire discussion preceding Eq. (23), the general solution for the displacement of the string in a single mode, is independent of the boundary conditions. It was only in the discussion following Eq. (23), that we specialized the solution to the case of the string fixed at $z = 0$ and L.

At a free end of a vibrating string, there is (by definition) no transverse force exerted on the end of the string, i.e., the frictionless rod exerts no transverse force on the frictionless ring. Then (by Newton's third law) the string and frictionless ring exert no transverse force on the frictionless rod. That means the string must be horizontal. *The slope of the string at a free end is zero at all times.* If one tries to exert a transverse force on the free end of a string, the string moves in such a way as to reduce the force to zero even as it is being applied. It never becomes different from zero, and the string remains horizontal, but of course not motionless. (The moral is that you cannot push on something that refuses to push back, but you can move it where you please.)

In Fig. 2.9 we show the modes of a string with one end fixed and the other free. We have labeled the successive modes according to the number of quarter-wavelengths contained in the string length L. Notice that the even harmonics with frequencies $2\nu_1$, $4\nu_1$, etc. are missing. The Fourier analysis of functions $f(z)$ with value zero at $z = 0$ and slope zero at $x = L$ is discussed in Prob. 2.29.

Dependence of tone quality on method of excitation. When a piano string is struck by its hammer, the fundamental (ν_1), the second harmonic or octave ($2\nu_1$), the octave plus a fifth ($3\nu_1$), the second octave ($4\nu_1$), the second octave plus a major third ($5\nu_1$), and the second octave plus a fifth ($6\nu_1$) are all excited to some extent, as are higher harmonics of the fundamental tone ν_1. The amount and phase of each Fourier component (each harmonic) depend on the initial configuration and velocity of all parts of the string at the instant just after it has been struck by the hammer. These depend to a great extent on the location of the hammer, i.e., on its distance from the end of the string. No mode that has a node (a permanently motionless point) at the striking point will be excited by the hammer blow, since the hammer imparts an initial velocity to the part of the string it hits. For example, if the string is plucked at its center, the modes with a node at the center are not excited. Inspection of Fig. 2.3 shows that in that case all the even harmonics are missing. Thus if we pluck the string for C128 in the middle, we expect it to vibrate in a superposition of C128, G384, E640, etc. The "tone quality" is then appreciably different from that produced when the string is struck near one end and vibrates in a superposition of C128, C256, G384, C512, E640, G768, etc.

Modes of homogeneous string form complete set of functions. Starting with a string fixed at both ends, we discovered that *any* reasonable function $f(z)$ that is defined between $z = 0$ and $z = L$ and that is zero at $z = 0$ and L can be expanded in the Fourier series

$$f(z) = \sum_{n=1}^{\infty} A_n \sin nk_1 z; \qquad k_1 L = \pi. \tag{54}$$

For that reason, the functions $\sin nk_1 z$, with $n = 1, 2, 3, \ldots$, are said to be a *complete set* of functions [with respect to functions $f(z)$ that vanish at $z = 0$ and L]. A complete set of functions is defined as a set such that *any* (reasonable) function $f(z)$ can be written as a superposition of functions from the set by choosing suitable constant coefficients.

Inhomogeneous string. Besides the sinusoidal functions that constitute a Fourier series, are there other complete sets? Yes, infinitely many sets! We can see this as follows. Suppose that the string is not homogeneous, i.e., that either its mass density or its tension (or both) is a continuous function of position z. (An example of a "string" with varying density and tension is provided by a vertically hanging slinky with fixed top and bottom ends. The tension at the bottom is less than that at the top by the weight Mg, where M is the total mass of the slinky.) Then the equation of motion of a small segment of string does not again lead to the classical

wave equation, which is

$$\frac{\partial^2\psi(z,t)}{\partial t^2} = \frac{T_0}{\rho_0}\frac{\partial^2\psi(z,t)}{\partial z^2}.$$

Instead, if we have equilibrium tension $T_0(z)$ and density $\rho_0(z)$, we easily find (Prob. 2.10) that we have

$$\frac{\partial^2\psi(z,t)}{\partial t^2} = \frac{1}{\rho_0(z)}\frac{\partial}{\partial z}\left[T_0(z)\frac{\partial\psi(z,t)}{\partial z}\right], \tag{55}$$

which reduces to the classical wave equation only if $T_0(z)$ and $\rho_0(z)$ are constants, independent of z. In a normal mode of this *inhomogeneous string*, just as in a mode of the homogeneous string, every part of the string vibrates in harmonic motion with the same frequency and phase constant:

$$\psi(z,t) = A(z)\cos(\omega t + \varphi). \tag{56}$$

Thus

$$\frac{\partial^2\psi}{\partial t^2} = -\omega^2 A(z)\cos(\omega t + \varphi), \tag{57}$$

$$\frac{\partial\psi}{\partial z} = \cos(\omega t + \varphi)\frac{dA(z)}{dz}. \tag{58}$$

Substituting these in Eq. (55) and canceling the common factor $\cos(\omega t + \varphi)$ yields the equation for the shape of the mode:

$$\frac{1}{\rho_0(z)}\frac{d}{dz}\left[T_0(z)\frac{dA(z)}{dz}\right] = -\omega^2 A(z). \tag{59}$$

Sinusoidal shape of standing waves is characteristic of homogeneous systems. The shape of the mode is given by $A(z)$, which is obtained by solving the differential equation Eq. (59) with the appropriate boundary conditions that $A(z) = 0$ at $z = 0$ and L. The function $A(z)$ is not sinusoidal in shape unless T_0 and ρ_0 are constants. Thus *sinusoidal* oscillations in space are only characteristic of the shapes of the normal modes of a *homogeneous* system.

Modes of inhomogeneous string form complete set of functions. We shall tell you without proof the characteristics of the normal modes for an inhomogeneous string with ends fixed at $z = 0$ and L. The lowest mode corresponds to a solution of Eq. (59), $A_1(z)$, which is zero only at $z = 0$ and L. (That is like one half-wavelength of a "distorted sine wave," which has no nodes between 0 and L.) This mode has frequency ω_1. The next mode has *one node* between $z = 0$ and L and thus resembles one full wavelength of a distorted sine wave. It has characteristic frequency ω_2. The mth mode has $m - 1$ nodes between $z = 0$ and L and resembles m half-wavelengths of a distorted sine wave. There are an infinite number

of modes (for a continuous string). The functions $A_1(z)$, $A_2(z)$, $A_3(z)$, . . . , which give the space dependence of the modes, form a complete set with respect to any reasonable function $f(z)$ that vanishes at $z = 0$ and L. A reasonable function $f(z)$ is defined to be one which the string or slinky can follow without violating any of our assumptions. In that case we can make a template that has the shape $f(z)$, fit the inhomogeneous string to the template, and let it go from rest at $t = 0$. The string will vibrate in an infinite superposition of its modes:

$$\psi(z,t) = \sum_{m=1}^{\infty} c_m A_m(z) \cos \omega_m t \tag{60}$$

Then at $t = 0$ we have

$$\psi(z,0) = f(z) = \sum_{m=1}^{\infty} c_m A_m(z). \tag{61}$$

Equation (61) shows that $f(z)$ (subject to our assumptions) can be expanded in the set of functions $A_m(z)$. Thus $A_m(z)$ form a complete set of functions. This argument is exactly analogous to the one that convinced us that the sinusoidal functions of a Fourier series form a complete set with respect to functions $f(z)$ that vanish at $z = 0$ and L.

Eigenfunctions. There are an infinite number of different ways that we can construct a string with nonuniform mass density and tension. Therefore, there are an infinite number of different complete sets $A_m(z)$. Sinusoidal functions of z are thus not the only complete set of functions for expanding functions $f(z)$. But they are a very important set, because they are very simple and easy to understand. Furthermore, they give the shapes of the modes whenever we have a system that is spatially *homogeneous.* When that is not the case, the sinusoidal functions are not very useful. Instead one tries to find and use the appropriate functions $A_m(z)$ that correspond to the normal modes of the system. These functions $A_m(z)$, or, more generally, $A_m(x,y,z)$ for a three-dimensional system, are called *eigenfunctions.* They give the *space* dependence of the normal modes.

For every position x, y, z, the *time* dependence of a mode is *always* given by $\cos(\omega t + \varphi)$. Thus a mode is essentially nothing but the simultaneous small oscillation (small enough to give linear equations) of all the moving parts, all parts oscillating with the same frequency and same phase constant. When the entire system is in a single mode, it pulsates and throbs like one big oscillator. Each mode has its own "shape," i.e., its own eigenfunction. The relation between mode frequency and shape is called the dispersion relation, $\omega(k)$, when the shapes of the eigenfunctions are sinusoidal. When they are not sinusoidal, there is, of course, no such thing as wavelength or wavenumber k. Then the relation between mode frequency and shape is not usually called by the name "dispersion relation."

We will not study inhomogeneous systems further. When you study quantum physics, you will study the eigenfunctions (shapes) of de Broglie standing waves in systems with nonconstant potential energy. These are analogous to the standing waves of an inhomogeneous string. See Supplementary Topic 2.

2.4 *Modes of a Noncontinuous System with N Degrees of Freedom*

In Sec. 2.2 we considered a continuous string, which is a system with infinitely many degrees of freedom. No real mechanical system has an infinite number of degrees of freedom, and we are interested in real systems. In this section we will find the exact solution for the modes of a uniform beaded string having N beads and with fixed ends. In the limit that we take the number of beads N to be infinite (and maintain the finite length L), we shall find the standing waves that we studied in Sec. 2.2. Our purpose is not merely that, however. Rather, we shall find that, in going to the limit of a continuous string, we discarded some extremely interesting behavior of the system. Remember that in order to use the smooth function $\psi(z,t)$ to describe the displacement when N is huge but not infinite, we have to prohibit ourselves from considering the highest modes, i.e., the modes $m = N, N - 1, N - 2$, etc. We must confine ourselves to values of m much less than N. That is because mode N has the zigzag configuration shown in Fig. 2.1, and thus neighboring beads do *not* have nearly the same displacement.

The most interesting new result we shall obtain in this section is that the dispersion law which we obtained for the *continuous* string, namely "ω equals a constant times k," does not generally hold. This relationship between frequency and wavelength, which implies that the frequency doubles when the wavelength is halved (i.e., which gives the harmonic frequency ratios), is an approximation that holds for the flexible string only in the continuous limit. The fact that it does not hold for a "lumpy" (but otherwise uniform) string provides an example of an interesting physical phenomenon called *dispersion*. A medium that satisfies the simple dispersion relation above, "ω equals constant times k," is called *nondispersive* (for the appropriate waves). If any other dispersion relation holds, the medium is called *dispersive*. Now consider the example:

Example 1: Transverse oscillations of beaded string

The system is shown in Fig. 2.10. There are N beads. The string is fixed at $z = 0$ and L. The beads are located at $z = a, 2a, \ldots, Na$. The total length L is $(N + 1)a$. The beads each have mass M. The string (or spring) segments are identical. They are massless and are perfect Hooke's law springs. The equilibrium tension is T_0. If the springs (strings) satisfy the slinky approximation (tension proportional to length), the oscillations

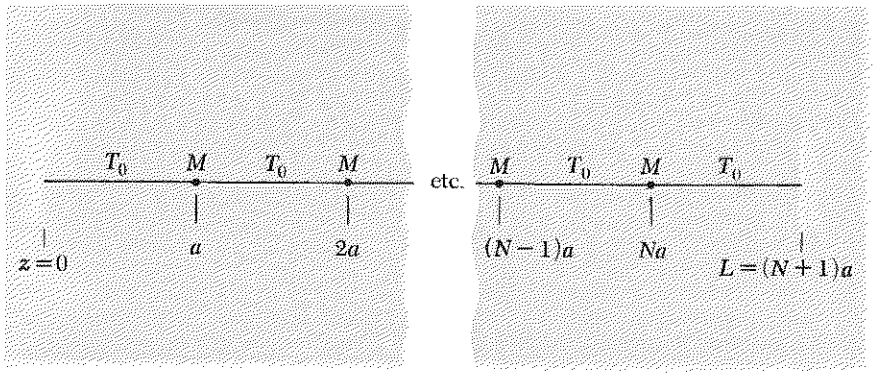


Fig. 2.10 Equilibrium configuration of beaded string.

may have arbitrarily large amplitude and still give us linear equations of motion. If the springs are not slinkies, we shall confine ourselves to small-amplitude oscillations, so as to get linear equations.

Now we consider the general configuration shown in Fig. 2.11. [It is not completely general, in that we are now considering only transverse oscillations along x. Later we shall consider the longitudinal oscillations along z. The general motion is of course a superposition of longitudinal oscillations along z and transverse oscillations along x and y.] The displacement of bead n upward (in the figure) from its equilibrium position is $\psi_n(t)$, with $n = 1, 2, 3, \ldots, N-1, N$. We focus our attention on an arbitrary bead n and its neighbors $n-1$ (to the left) and $n+1$ (to the right).

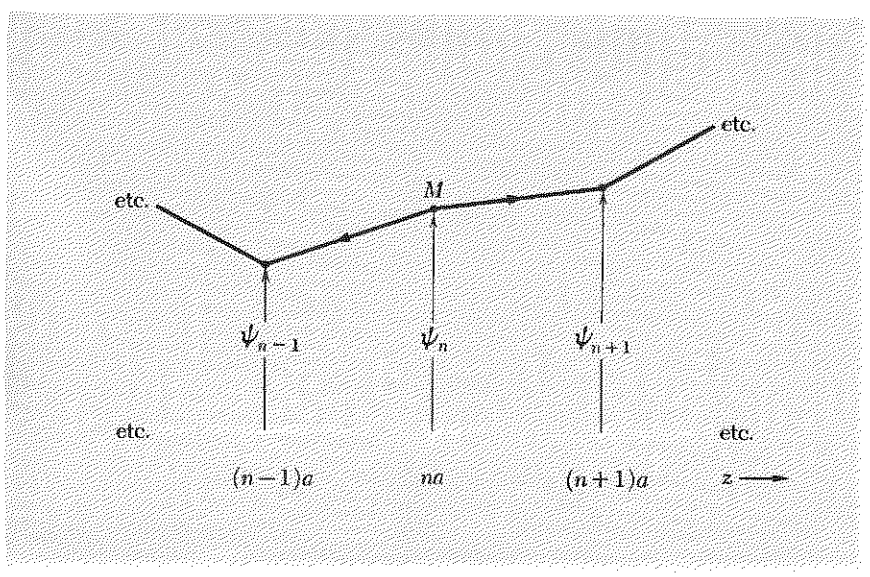


Fig. 2.11 General configuration of beaded string for transverse oscillations along x.

Equation of motion. We want the equation of motion for bead n. We have already solved a problem very similar to this one (in Sec. 1.2, for one degree of freedom, and in Sec. 1.4 for two degrees of freedom). Therefore, we leave it to you to show that, for either the slinky approximation or the small-oscillations approximation, Newton's law applied to the motion of bead n gives

$$M\frac{d^2\psi_n(t)}{dt^2} = T_0\left[\frac{\psi_{n+1}(t) - \psi_n(t)}{a}\right] - T_0\left[\frac{\psi_n(t) - \psi_{n-1}(t)}{a}\right]. \tag{62}$$

Equation (62) is perfectly general; it holds for an arbitrary motion of the freely oscillating system, i.e., for an arbitrary superposition of the N different modes.

Normal modes. We want to find the frequencies and configurations of the individual modes. Therefore, we assume that we have a single mode of frequency ω. Each bead oscillates harmonically with the same frequency ω and with the same phase constant φ. The shape of the mode is given by the ratios of amplitudes of the beads. We let A_n designate the vibration amplitude for bead n in the mode we are considering. Thus we have, in a single mode,

$$\begin{aligned}
\psi_1(t) &= A_1 \cos(\omega t + \varphi);\ \psi_2(t) = A_2 \cos(\omega t + \varphi);\ \ldots;\\
\psi_{n-1}(t) &= A_{n-1} \cos(\omega t + \varphi);\ \psi_n(t) = A_n \cos(\omega t + \varphi);\\
\psi_{n+1}(t) &= A_{n+1} \cos(\omega t + \varphi);\ \ldots.
\end{aligned} \tag{63}$$

Because of Eq. (63), we have

$$\frac{d^2\psi_n(t)}{dt^2} = -\omega^2\psi_n(t) = -\omega^2 A_n \cos(\omega t + \varphi). \tag{64}$$

Now use Eq. (64) on the left side of Eq. (62), and Eqs. (63) on the right side. Then cancel the common time-dependent factor $\cos(\omega t + \varphi)$. This gives

$$-M\omega^2 A_n = \frac{T_0}{a}(A_{n+1} - 2A_n + A_{n-1});$$

i.e.,

$$A_{n+1} + A_{n-1} = A_n\left(2 - \frac{Ma}{T_0}\omega^2\right). \tag{65}$$

Equation (65) looks formidable. It gives the shape of the mode with angular frequency ω. Let us solve it by bold guesswork. We are guided in our guessing by our previous solution for the modes of a continuous

string with fixed ends at $z = 0$ and L. In that problem we found the shape of the modes to be given by

$$A(z) = A \sin \frac{2\pi z}{\lambda} = A \sin kz. \tag{66}$$

Our solution for A_n must of course reduce to Eq. (66) in the limit of infinitely many beads (the continuous limit). Let us try a solution obtained by simply setting $z = na$ in Eq. (66):

$$A_n = A \sin \frac{2\pi na}{\lambda} = A \sin kna. \tag{67}$$

Then

$$\begin{aligned} A_{n+1} &= A \sin k(n+1)a = A \sin (kna + ka) \\ &= A(\sin kna \cos ka + \cos kna \sin ka). \\ A_{n-1} &= A \sin k(n-1)a = A \sin (kna - ka) \\ &= A(\sin kna \cos ka - \cos kna \sin ka). \\ A_{n+1} + A_{n-1} &= 2A \sin kna \cos ka = 2A_n \cos ka \end{aligned} \tag{68}$$

Inserting Eq. (68) into Eq. (65), we obtain

$$2A_n \cos ka = A_n\left(2 - \frac{Ma}{T_0}\omega^2\right). \tag{69}$$

Exact dispersion relation for beaded string. Equation (69) is supposed to hold for every bead n, whether or not A_n happens to be zero for that particular bead n in a particular mode. Therefore, we can take n to correspond to a bead that is not at a node, i.e., one for which A_n is not zero. Then we cancel A_n and obtain the condition that our guess for a solution must meet to be an actual solution:

$$2 \cos ka = 2 - \frac{Ma}{T_0}\omega^2,$$

i.e.,

$$\begin{aligned} \omega^2 &= \frac{2T_0}{Ma}(1 - \cos ka) \\ &= \frac{2T_0}{Ma}\left[1 - \left(\cos^2 \frac{ka}{2} - \sin^2 \frac{ka}{2}\right)\right], \\ \omega^2 &= \frac{4T_0}{Ma} \sin^2 \frac{ka}{2} = \frac{4T_0}{Ma} \sin^2 \frac{\pi a}{\lambda}. \end{aligned} \tag{70}$$

Equation (70), which relates frequency to wavelength (or wavenumber) for a mode with angular frequency ω, is the dispersion relation for the beaded string.

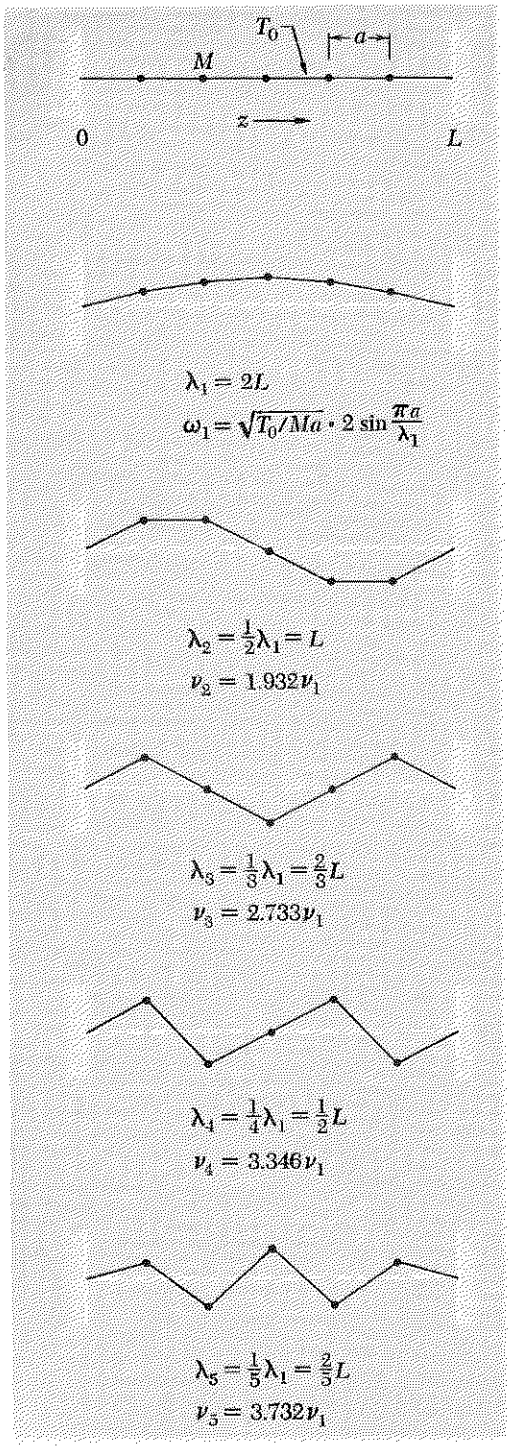


Fig. 2.12 ***Modes of a string with five beads.***

Boundary conditions. We have not yet specified the boundary conditions completely. When we wrote Eq. (67) instead of the more general expression

$$A_n = A \sin kna + B \cos kna, \tag{71}$$

we had already satisfied the boundary condition at $z = 0$, namely that the string displacement be zero there for any mode. Setting $z = na = 0$ in Eq. (71) and demanding $A_0 = 0$ gives $B = 0$. We must still satisfy the boundary condition at $z = L$, namely that the string displacement be zero there also. The wall at $z = L$ corresponds to the "fixed bead $N + 1$." Thus we need $A_{N+1} = 0$:

$$A_{N+1} = A \sin k(N + 1)a = A \sin kL = 0. \tag{72}$$

There are N possible solutions for Eq. (72). Each solution corresponds to a single mode m, with $m = 1, 2, \ldots, N$. We number the modes so that $m = 1$ has the longest wavelength. Thus we have

$$k_1L = \pi, \quad k_2L = 2\pi, \quad \cdots, \quad k_mL = m\pi, \quad \cdots, \quad k_NL = N\pi. \tag{73}$$

The reason that there are only N solutions [the modes specified by Eq. (73)] is that the last term in the sequence of Eq. (73) corresponds to a completely zigzag configuration: starting at $z = 0$, the first string segment "zigs" up to the first bead, the second string "zags" down to bead 2, . . . , string $N + 1$ zags (or zigs) from bead N to the wall. Equation (72) does have the further solutions $k_{N+1}L = (N + 1)\pi$, $k_{N+2}L = (N + 2)\pi$, etc., but to do all the zigzagging such solutions imply, takes more string segments than we have.

The equation for the shape of the modes, Eq. (65), was obtained without consideration of boundary conditions. (Figure 2.11 contains no boundary conditions.) The most general solution of this equation is given by Eq. (71), with B/A and k determined by the boundary conditions. If you substitute Eq. (71) into Eq. (65), you find the dispersion relation Eq. (70), independent of the boundary conditions, i.e., independent of the values of A, B, and k, as you can easily show (Prob. 2.19). For our particular boundary conditions (string fixed at $z = 0$ and L), we obtain the mode configurations of Eq. (72) with k_m given by Eq. (73). Then the frequencies ω_m are given by Eq. (70).

Notice that the mode configurations of Eq. (73) are exactly the same as those that we obtained for the continuous string, the only difference being that for the continuous string we had $N = \infty$ and thus no highest mode. Also, for the beaded string, the string segments are straight and do not follow the smooth sinusoidal function A_m that passes through the beads.

We illustrate the modes by showing the case $N = 5$ in Fig. 2.12. In Fig. 2.13 we plot the dispersion relation given by Eq. (70):

$$\omega(k) = 2\sqrt{\frac{T_0}{Ma}} \sin \frac{ka}{2}. \tag{74}$$

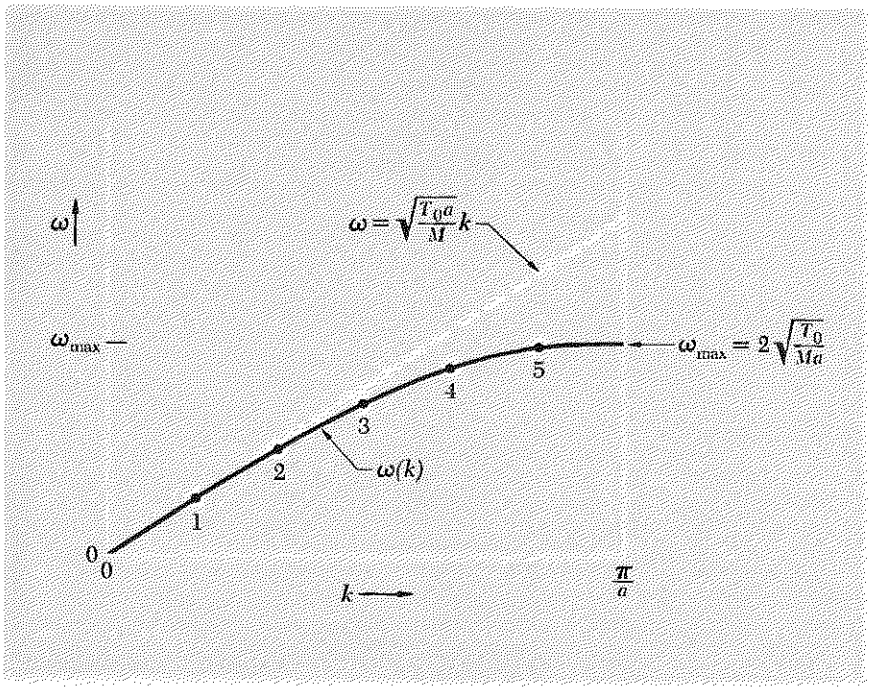


Fig. 2.13 Dispersion relation for the beaded string. The five labeled points correspond to the five modes of a string of five beads with both ends fixed. Different boundary conditions or a different number of beads would give different points on the same diagram.

The five labeled points give k and ω for the five modes of the string of five beads with both ends fixed. If there were a different number of beads or different boundary conditions (for example we could have a free end at $z = L$), the points representing the modes would lie at different places on the *same curve* $\omega(k)$. Thus Fig. 2.13 holds for *every* beaded string.

Continuous or long wavelength limit. In the continuous approximation, we assume an infinite number of beads in the region between $z = 0$ and $z = L$. Thus the bead separation a goes to zero. It is interesting to look at the properties of our exact dispersion relation, Eq. (74), for bead separations a that are "very small" but not *exactly* zero, in order to see that the dispersion relation approaches that of the continuous string. We have to know what we mean by "small", i.e., small with respect to what? The continuous approximation is a good one when bead spacing a is small compared with the wavelength λ:

$$a \ll \lambda; \qquad ka = 2\pi\frac{a}{\lambda} \ll 1.$$

We now use the Taylor's series expansion [Appendix Eq. (4)],

$$\sin x = x - \tfrac{1}{6}x^3 + \cdots.$$

We insert this series with x equal to $\frac{1}{2}ka$ into Eq. (74):

$$\omega(k) = 2\sqrt{\frac{T_0}{Ma}}\left[\frac{1}{2}ka - \frac{1}{48}(ka)^3 + \cdots\right]$$
$$= \sqrt{\frac{T_0 a}{M}}k\left[1 - \frac{1}{24}(ka)^2 \cdots\right],$$

i.e.

$$\omega(k) \approx \sqrt{\frac{T_0 a}{M}}k. \tag{75}$$

Equation (75) is the "nondispersive" dispersion law that we obtained for the continuous string in Sec. 2.3, with $M/a = \rho_0$.

Dispersion relation for real piano string. We have discovered that the modes of a noncontinuous string do not satisfy the nondispersive wave dispersion relation of Eq. (75). Therefore, we expect that the overtones of a piano string, for example, the string with fundamental tone C128, are not given exactly by the octave C256, twelfth G384, double octave C512, etc. That is correct; they are not. According to Eq. (74) or, more easily, its plot in Fig. 2.13, an increase in k does not produce a proportional increase in frequency, but rather slightly less. Therefore, you might expect the overtones of the piano string to be slightly "flat" with respect to the predictions of the continuous-string theory, i.e., you might expect the second harmonic to have $\nu_2 < 256$, the third to have $\nu_3 < 384$, etc. That is incorrect! The overtones of a piano string are not flat; they are *sharp* with respect to the simple "harmonic-overtone" prediction of Eq. (75)! That is because, although the model of a perfectly continuous and perfectly flexible string is not a perfect description of a piano string, neither is the beaded-string model. In fact the beaded-string model is worse than the continuous-string model, since it gives to the result of the latter a "correction" with the wrong sign! The trouble with the continuous-string model for a piano string is not that it needs some beads, but rather that a real piano string is not perfectly flexible. When you bend it, it wants to straighten out again, even if there is no tension helping to pull it straight. Consequently, the return force on a small curved segment of string (i.e., the force that tends to straighten the string out—it is straight at equilibrium) is slightly greater than that predicted by the "completely flexible" model. The mode frequency is of course given by ω^2 = return force per unit displacement per unit mass. The higher modes have shorter wavelengths, so they are bent more. The stiffness is thus more important for the higher modes than for the lower, and hence the frequency increases *faster* than expected from the flexible-string model.

There is an interesting "fine point" to this explanation. The return force due to the tension and that due to the stiffness *both* increase with k.

Therefore, if the stiffness is to play a relatively larger role at higher values of k than at its lower values, the return force due to stiffness must increase with k at a rate greater than the return force due to tension. The return force due to tension is proportional to k^2. That due to stiffness turns out to be proportional to k^4. Thus the dispersion relation for a real piano string is given by

$$\omega^2 \approx \frac{T_0}{\rho_0} k^2 + \alpha k^4, \tag{76}$$

where α is a positive constant that is due to the stiffness. If the stiffness term were also proportional to k^2, we would still get the "nondispersive" dispersion relation, Eq. (75), with T_0/ρ_0 merely replaced by $(T_0/\rho_0) + \alpha$. Then the frequency ratios would still be the "harmonic" ones $\nu_2 = 2\nu_1$, $\nu_3 = 3\nu_1$, etc. Now we shall consider more examples:

Example 2: Longitudinal oscillations of a system of springs and masses

This is an important example because it will later provide us with a very simple model to help us understand sound waves. (Sound waves consist of longitudinal vibrations; i.e., the vibrations are perpendicular to the "wavefronts".)

We have already studied the cases $N = 1$ and 2 in Secs. 1.2 and 1.4, respectively. We now consider the general case of N masses coupled by springs as shown in Fig. 2.14.

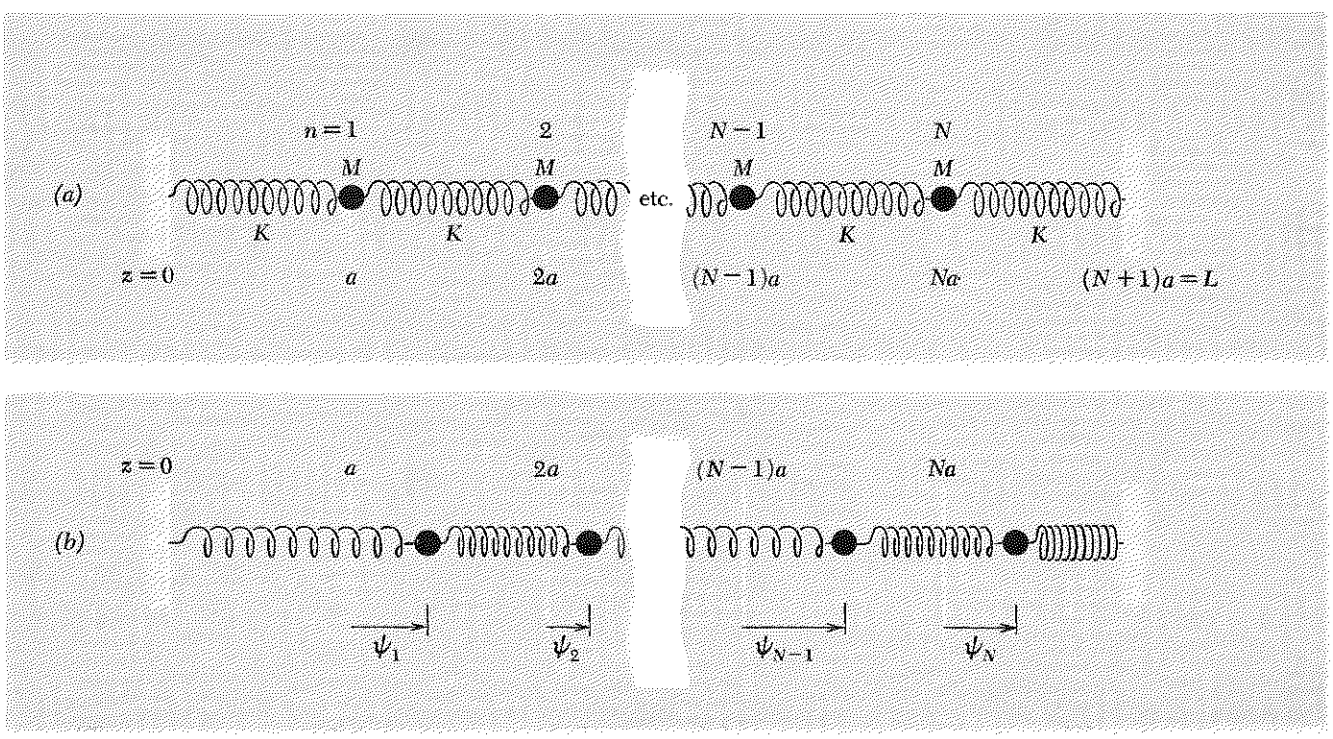


Fig. 2.14 Longitudinal oscillations of N masses and $N + 1$ springs. (a) Equilibrium configuration. (b) General configuration.

The equation of motion of bead n is very easily derived. (If you have difficulty, review the derivation for $N = 2$ in Sec. 1.4.) One finds

$$M\frac{d^2\psi_n}{dt^2} = K(\psi_{n+1} - \psi_n) - K(\psi_n - \psi_{n-1}). \tag{77}$$

The mathematical form of Eq. (77) is the same as that of the equation of motion for transverse displacements, Eq. (62), except for the replacement of constant T_0/a by the spring constant K. Therefore, all of our former mathematical steps can be repeated. Thus we get the dispersion relationship [obtained by replacing T_0/a by K in Eq. (74)]

$$\omega(k) = 2\sqrt{K/M}\sin\frac{ka}{2} = 2\sqrt{K/M}\sin\frac{\pi a}{\lambda}. \tag{78}$$

In the mode with wavenumber k, the motion of mass n is given by

$$\psi_n(t) = A\sin nka\cos[\omega(k)t + \varphi], \tag{79}$$

with the N different possibilities for k given by

$$k_1L = \pi, \qquad k_2L = 2\pi, \qquad \cdots, \qquad k_NL = N\pi. \tag{80}$$

The dispersion relation plotted in Fig. 2.13 need only be suitably relabeled to represent Eq. (78).

Lumped parameters and distributed parameters. When we considered the transverse vibrations of a beaded string, we went to the continuous limit by letting the bead spacing a go to zero (keeping L constant). Once one has gone to sufficiently small a/λ, so that the continuous approximation is a good one, then one can use a different physical model of the system. Instead of letting the spacing a continue to go to zero while maintaining the mental picture of massless springs alternating with point masses, we can distribute the mass uniformly along the spring. Then there are no longer lumped masses and massless springs (lumped spring constants). Instead there is just one long spring with the mass distributed along it. A good example is provided by a slinky. For the "repeat length" a we can now take the length along z of a single turn of the helical spring. The meanings of the parameters M and K become respectively the mass of one turn (of the helix) and the spring constant of one turn. If one has a total of N turns (N is not the number of degrees of freedom, now), then the total mass is NM. The total spring constant (i.e., the spring constant of the complete spring of length $L = Na$) is K/N. (That is because two identical springs hooked in series make one longer spring with half the spring constant of the two component springs.)

Instead of retaining the repeat length a (of one turn of the helix), we can eliminate it completely from the notation (in the continuous approxima-

tion) by replacing M/a by the mass per unit length (linear mass density) $\rho_0 = M/a$. Similarly, one can eliminate K, the spring constant of one turn, replacing it by a quantity that is characteristic of the material of the spring and its construction. That quantity is the inverse spring constant per unit length, K^{-1}/a. That is easily seen as follows. For a spring of total length $L = Na$, the spring constant K_L is N times smaller than K:

$$K_L = \frac{1}{N}K = \frac{a}{L}K. \tag{81}$$

Thus we have $K_L \cdot L = Ka$, which is independent of L; i.e., Ka is a property of the "springiness" of the material and is independent of the length of the spring. Since we like to deal with quantities that have dimensions of "something per unit length," we write the relation

$$K_L \cdot L = Ka$$

in the form

$$\frac{K^{-1}}{a} = \frac{K_L{}^{-1}}{L}. \tag{82}$$

Now we can express the result by saying that the inverse spring constant per unit length is a property of the spring which is independent of its length.

Example 3: Slinky

A slinky is a helical spring having $N \approx 100$ turns, with each turn about 7 cm in diameter. The unstretched length is about 6 cm. When stretched to a length L of about a meter, it satisfies the slinky approximation very well. A convenient "repeat length" a is given by the length per turn, $a = L/N$. Then K is the spring constant for one turn, and K^{-1}/a is independent of the length L. (The mass is of course distributed, not lumped at intervals of length a.) The dispersion relation for longitudinal oscillations is obtained by going to the continuous limit, starting with Eq. (78):

$$\begin{aligned}
\omega(k) &= 2\sqrt{\frac{K}{M}}\sin\frac{ka}{2} \\
&= 2\sqrt{\frac{K}{M}}\left[\frac{ka}{2} - \frac{1}{6}\left(\frac{ka}{2}\right)^3 + \cdots\right] \\
&\approx \sqrt{\frac{Ka^2}{M}}k \\
&= \sqrt{\frac{Ka}{(M/a)}}k.
\end{aligned} \tag{83}$$

The dispersion relation for transverse oscillations is [see Eq. (75)]

$$\begin{aligned}\omega(k) &\approx \sqrt{\frac{T_0}{M/a}}\,k \\ &\approx \sqrt{\frac{Ka}{M/a}}\,k,\end{aligned} \tag{84}$$

since $T_0 = Ka$ in the slinky approximation. Thus the slinky has the same dispersion relation for longitudinal and for transverse oscillations. Therefore, *if* the boundary conditions are the same (for example, both ends rigidly fixed for vibrations along x, y, or z), the modes for x, y, and z vibrations have the same sequence of wavelengths and frequencies. You can easily verify for yourself that the longitudinal and transverse modes have the same frequencies. We strongly recommend that you now perform some of the Home Experiments involving slinkies. There is no better way to understand waves than to make your own. Get a slinky. (Made by James Industries, Inc., Hollidaysburg, Pa.; available in any toy store for about $1.)

Example 4: *LC* network

Consider the sequence of coupled inductances and capacitances shown in Fig. 2.15. From Fig. 2.15*b* (and our discussion involving the same network for the case $N = 2$, in Sec. 1.4), we easily find that the equation for the electromotive force across the nth inductance is given by

$$L\frac{dI_n}{dt} = -C^{-1}Q' + C^{-1}Q.$$

Then

$$L\frac{d^2I_n}{dt^2} = -C^{-1}\frac{dQ'}{dt} + C^{-1}\frac{dQ}{dt}.$$

Using conservation of charge to eliminate dQ'/dt and dQ/dt, we get

$$\begin{aligned}L\frac{d^2I_n}{dt^2} &= -C^{-1}[I_n - I_{n+1}] + C^{-1}[I_{n-1} - I_n] \\ &= C^{-1}[I_{n+1} - I_n] - C^{-1}[I_n - I_{n-1}].\end{aligned} \tag{85}$$

Equation (85) has the same mathematical form as Eq. (77), the equation of motion governing the longitudinal oscillations of a sequence of masses and springs. Therefore, without yet worrying about the boundary conditions, we can write down the dispersion relation and the general solution for the currents in the inductances. The dispersion relation is obtained by replacing K/M by C^{-1}/L in Eq. (78):

$$\omega(k) = 2\sqrt{\frac{C^{-1}}{L}}\sin\frac{ka}{2}. \tag{86}$$

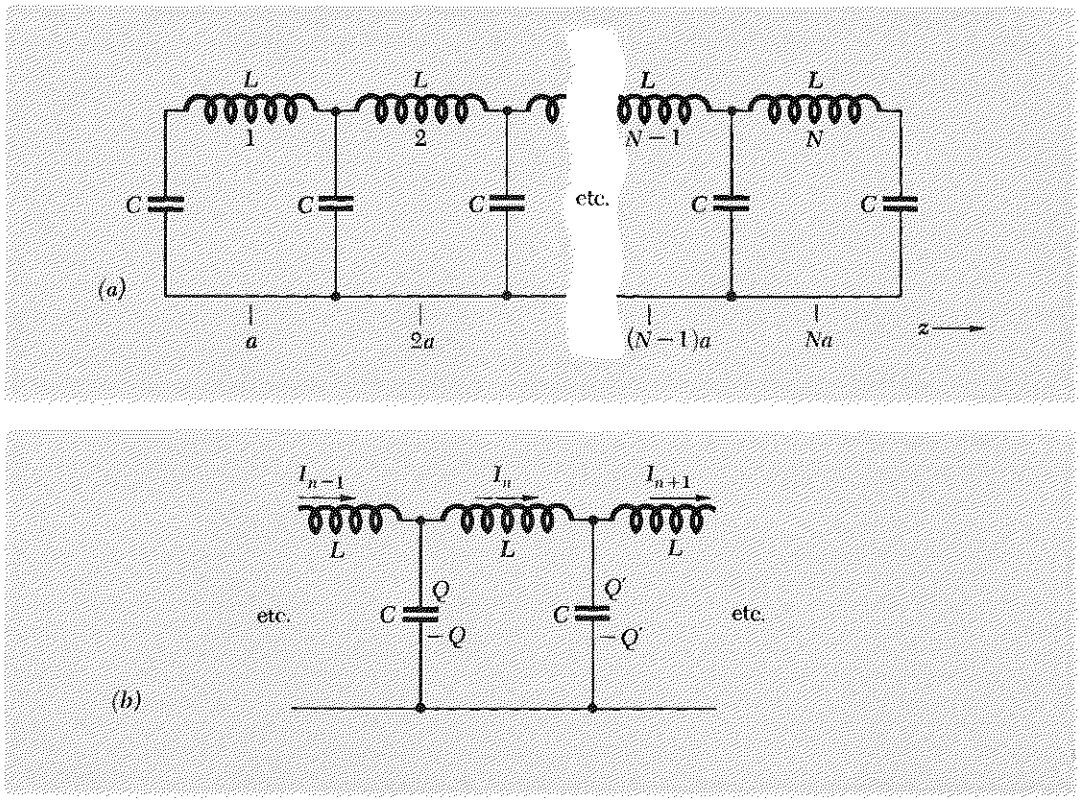


Fig. 2.15 Network of coupled inductances and capacitances. (a) The lumped parameters. (b) General current and charge configuration at nth inductance.

The general solution of Eq. (85) for a single mode, without regard to boundary conditions, is

$$I_n(t) = [A \sin nka + B \cos nka] \cos [\omega(k)t + \varphi], \tag{87}$$

where the constants A and B and the sequence of values of k corresponding to modes depend on the boundary conditions at the ends of the system.

The meaning of ka. You may have noticed that the equation governing the behavior of the LC circuit, Eq. (85), does not contain the distance a. We labeled such a distance in Fig. 2.15, but there was no need to do so, since a circuit diagram is not a spatial diagram, and the behavior of the circuit does not depend on its spatial configuration. What then do we mean by ka in the dispersion relation, Eq. (86), and in the general solution for the currents, Eq. (87)? When the length along z really has an important physical significance, as in the oscillating string, then we know that k means the increase per unit length along z of the phase of the function $A \sin kz + B \cos kz$ which gives the shape of the mode. When we have lumped parameters, as in the beaded string, we write $z = na$, where $n = 1, 2, \ldots$ is the bead index. Then, the quantity ka appearing in the

shape function $A \sin nka + B \cos nka$ is the product of radians of phase (of the shape function) per unit distance times the distance between lumped masses. Thus ka is the number of radians of phase increase from the lumped mass n to the succeeding lumped mass $n + 1$. In the case of the system of lumped inductances and capacitances, the quantity ka is similarly the increase in phase of the "shape" function $A \sin nka + B \sin nka$ when we go from one lumped inductance to the next. We do not really need to specify the distance a by which the inductances are separated. We could merely replace ka by some symbol, say θ, where θ would then denote the increase of phase when n is increased by 1 in the shape function $A \sin n\theta + B \cos n\theta$. That notation is too abstract for us and abandons the mathematical similarity to the mechanical examples, so we shall retain the idea that the lumped inductances are separated by distance a.

Other forms of dispersion relation. You may have noticed that every lumped-parameter system we have considered so far in this section has an exact dispersion relation of the form

$$\omega(k) = \omega_{\max} \sin \frac{ka}{2}, \tag{88}$$

as plotted in Fig. 2.13, with $\omega_{\max}$ a constant depending on the physical system. That is only because of our choice of systems. In every case we considered, we chose a system in which the return force on a given mass (or inductance) is entirely the result of the coupling of that mass to neighboring masses and is proportional to the relative displacement of that mass and its neighbors. Such systems are numerous, but there are many other interesting and important forms of dispersion relations. For example, there are systems having the property that the return force on a given moving part has the following two independent contributions. One is due to the force arising from coupling with similar neighboring moving parts. If this were the only contribution, the dispersion relation would take the form of Eq. (88). The other is due to its coupling to some "external" force. This external contribution depends only on the displacement of the moving part from its equilibrium position and not on the displacements of the neighboring parts. If it were the only contribution, then the moving parts would be uncoupled, and their displacements would be the normal coordinates of the entire system. This kind of system is illustrated by the following example.

Example 5: Coupled pendulums

The system is shown in Fig. 2.16. Each mass has a return force made up of two contributions. The "external" contribution is that due to gravity. It is proportional to the displacement of the mass from its equilibrium position and is independent of the displacement of its neighbors. The

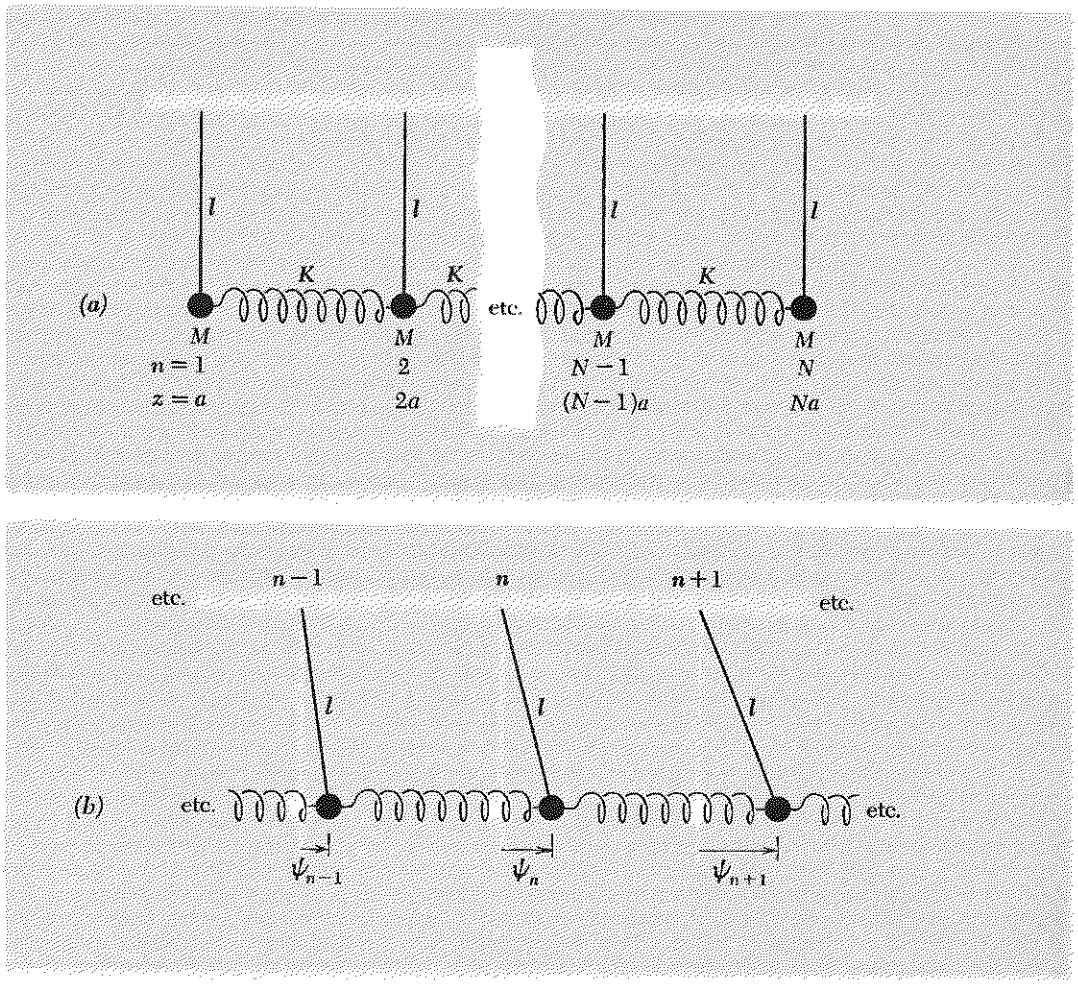


Fig. 2.16 ***Coupled pendulums. (a) Equilibrium. (b) General configuration.***

second independent force contribution is due to the coupling of a given mass to its neighbors by means of the springs. This contribution depends on the displacement of the neighboring masses.

Let us try to guess the dispersion relation. If we had only the coupling between masses, i.e., if g were zero, then we would have the dispersion relation for longitudinal oscillations of coupled masses. Thus the return force per unit displacement per unit mass, ω^2, would be given by Eq. (78):

$$\omega^2 = 4\frac{K}{M}\sin^2\frac{ka}{2}, \qquad \text{if } g = 0. \tag{89}$$

Now suppose that (with $g = 0$) we have oscillation in a single mode, with a definite shape determined by the definite value of k, which is set by the boundary conditions. Imagine that we can gradually increase g from zero

up to its final value of 980 (cgs units), using a "gravity knob." (A more practical method can be invented. What else could you vary?) When we increase g from zero to the very small value g', the return force per unit displacement per unit mass for *each* particle increases by the *same* amount, the contribution from g':

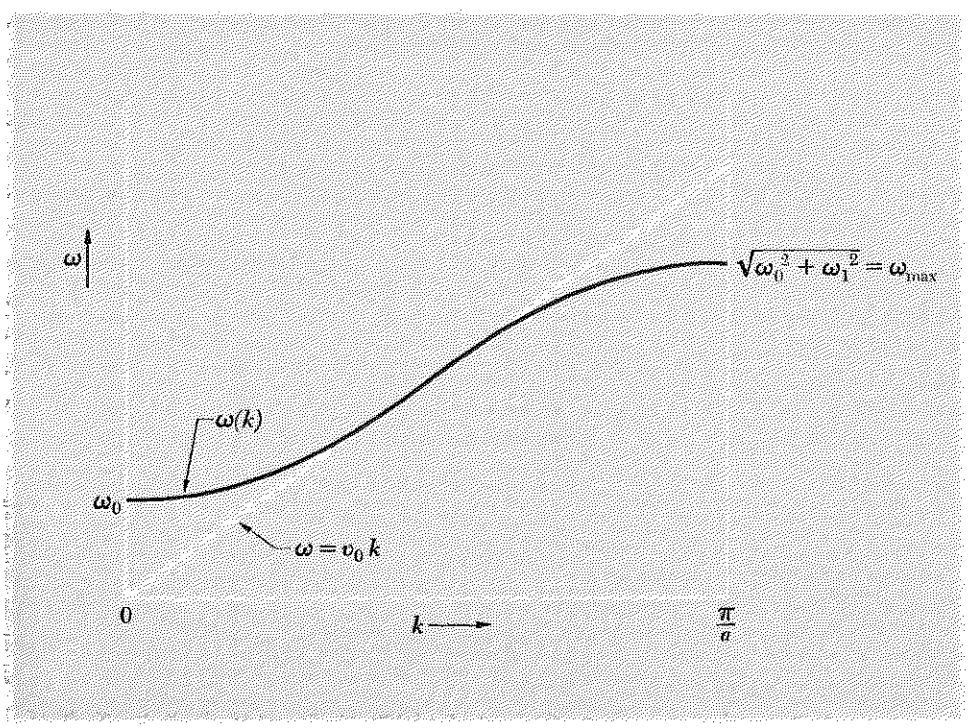
$$\text{Contribution of } g' \text{ to } \omega^2 \text{ is } \frac{g'}{l} \text{ for every mass.}$$

That means that the masses will continue to oscillate with the same configuration, same k, and same linear combination of $\sin kz$ and $\cos kz$, but they will merely oscillate faster. That is because they *had* the same return force per unit displacement per unit mass, ω^2, when g' was zero, and now we have added the *same amount* to the frequency squared of every mass. Therefore, all masses still have the same ω^2 and hence are still in a mode. Thus by gradually turning on g, we preserve the modes without mixing them up. The shapes and wavelengths are the same as for $g = 0$, and the total return force per unit mass per unit displacement is now

$$\omega^2(k) = \frac{g}{l} + \frac{4K}{M}\sin^2\frac{ka}{2}. \tag{90}$$

If you prefer a less qualitative derivation, of this relation, see Prob. 2.26; there *you* will find the equation of motion of mass n, verify the dispersion relation Eq. (90), and find the shapes of the modes. (Can you already see that for the boundary conditions of Fig. 2.16, the lowest mode has $k = 0$?)

Fig. 2.17 Dispersion relation for coupled pendulums.



Later we shall encounter more examples of dispersion laws of the form of Eq. (90), which we can write in the general form

$$\omega^2(k) = \omega_0^2 + \omega_1^2 \sin^2 \frac{ka}{2}. \tag{91}$$

In the continuous limit, where we have $ka \ll 1$, this relation becomes

$$\omega^2(k) = \omega_0^2 + v_0^2 k^2, \tag{92}$$

where v_0^2 is the constant $\omega_1^2 a^2/4$.

We shall encounter dispersion laws of the form of Eq. (92) when we study electromagnetic radiation in a waveguide and electromagnetic waves in the earth's ionosphere. (That is also the form of dispersion law for relativistic de Broglie waves in the quantum description of particles.) We plot Eq. (91) in Fig. 2.17.

Example 6: Plasma oscillations

This is an interesting example of a system with a dispersion relation like that of coupled pendulums. In Chap. 4 we shall derive the dispersion relation for electromagnetic waves in the earth's ionosphere. It turns out to have the form of Eq. (92):

$$\omega^2(k) = \omega_p^2 + c^2 k^2, \tag{93}$$

where c is the velocity of light and where ω_p, called the "plasma oscillation frequency," is given by

$$\omega_p^2 = \frac{4\pi N e^2}{m}. \tag{94}$$

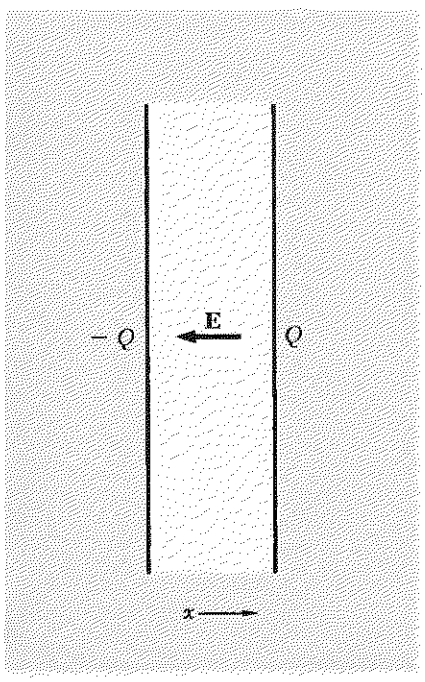
Here N is the number density of electrons (in electrons per cm^3), e is the charge on the electron, and m is the electron mass. From Fig. 2.17 we see that the mode of lowest possible frequency for a system with a dispersion relation like Eq. (91) or (92) is a mode with $k = 0$, i.e., with infinite wavelength. That just means all the pendulums oscillate with the same phase constant and same amplitude. The pendulum frequency is then given by $\omega^2 = g/l$. The frequency of that lowest mode in the present example is the plasma oscillation frequency ω_p, as we see by setting $k = 0$ in Eq. (93). We shall now consider that mode and derive Eq. (94) as its frequency.

A neutral plasma consists of a gas of neutral molecules with some of the molecules ionized. Every singly ionized molecule consists of one positive ion which has set free one (negative) electron. The earth's ionosphere is a layer of air (actually several layers with somewhat differing properties) that contains many ionized air molecules (N_2 and O_2 molecules). The ionization of an air molecule usually occurs by means of absorption of an ultraviolet-light quantum emitted by the sun. The ion and free-electron

density is greatest at about 200 to 400 km above the earth's surface. Higher than that, the electron (and ion) density decreases because the density of neutral air molecules available to be ionized decreases. Lower than that, the electron density decreases because the ultraviolet radiation has already been mostly absorbed. (We would soon die of sunburn without the protective air layer above us.)

Since the plasma is neutral (on the average), it does not act as a source of electrostatic field. However, one region of the plasma may have at any one instant a slight excess of charge, with a corresponding deficit in some other neighboring region. This creates an electric field in the plasma. Under the influence of the electric field, the ions are accelerated in one direction (i.e., along the field), and the electrons in the other. The charges move in directions that tend to cancel those charge excesses and deficits which created the electric field. Thus we have a "return force." By the time the charge excess has been canceled to zero and the corresponding electric field reduced to zero, the ions and electrons have acquired velocities. Their inertia causes them to "overshoot," and we get a new charge excess and deficit with opposite sign from the original one. We have here the typical situation that sustains oscillations, once they are excited.

Fig. 2.18 Oscillations in a confined plasma.



If we are interested only in the net motion of charge back and forth from one region to another, we can forget about the positive ions and consider the entire motion of charge to be due to the motion of the electrons. This is because the acceleration of an electron is larger than that of a singly charged ion by the ratio of their respective masses (about 3×10^4), since the electric force is the same on each.

Let us study a simplified situation where the plasma is contained between confining walls. We neglect motion of the ions compared with that of the electrons. At any instant there may be an excess of charge Q on one wall and a corresponding deficit on the other wall. This produces a spatially uniform electric field in the plasma (Vol. II, Sec. 3.5) given by

$$E_x = -4\pi \frac{Q}{A}, \tag{95}$$

where A is the area of the wall, and where the minus sign signifies that E_x tends to return Q to zero. There is no other source of electric field. (The plasma between the walls is neutral, because every electron that moves to the right out of a given region is replaced by another moving in from the left.) A single electron has mass m and charge q. Newton's second law gives

$$\frac{m\, d^2x}{dt^2} = qE_x \tag{96}$$

for every electron in the plasma. (We are neglecting other forces on the electrons which arise from collisions among the electrons and ions; these forces average to zero and give no net motion of charge.) Now suppose that there are N free electrons per cubic centimeter and that each electron is displaced from its average (equilibrium) position by a distance x. Then the net charge deposited on one wall (and removed from the other) is given by

$$Q = NqAx. \tag{97}$$

Differentiating Eq. (97) twice with respect to time and inserting Eqs. (96) and (95) yields

$$\frac{d^2Q}{dt^2} = -\frac{4\pi Nq^2}{m}Q. \tag{98}$$

This has the solution

$$Q = Q_0 \cos(\omega t + \varphi),$$

with

$$\omega^2 = \frac{4\pi Ne^2}{m} \equiv \omega_p{}^2. \tag{99}$$

The quantity ω_p is called the plasma oscillation frequency.

The free-electron density N in the earth's ionosphere varies with height and with time. The recombination of ions and electrons to form neutral molecules continues after sunset, but the forming of new ions ceases. The electron density therefore decreases at night. Typical daytime plasma oscillation frequencies $\nu_p(=\omega_p/2\pi)$ are

$$\nu_p = 10 \text{ to } 30 \text{ Mc (megacycles per second)}, \tag{100}$$

which corresponds to $N \approx 10^6$ to 10^7 free electrons per cm^3.

Esoteric examples

If one combines De Broglie's hypothesis, which says that a particle of momentum p has a wave number k given by $p = \hbar k$, with the "Bohr frequency condition," which says that a particle of energy E has a wave frequency ω given by $E = \hbar\omega$, one can then find a dispersion relation between ω and k for particles, given the relation between E and p. Examples are given in Supplementary Topic 2.

Problems and Home Experiments

Home experiment

2.1 Slinky—dependence of frequency on length. Hold the first turn of your slinky in your left hand and the last turn in your right, with your hands about 3 ft apart. Measure the frequency for vertical transverse oscillations. (Don't worry about the sag.) Next stretch the slinky out as far as you can reach. Measure the frequency. Now fasten each end to something so that the total length is 8 or 10 feet. Measure the frequency. Explain your result. Use your frequency measurement to determine the inverse spring constant per turn. Suppose that N_0 is the total number of turns of the slinky. Hold or fasten the slinky so that only N turns are free. Before doing the experiment, predict the frequency dependence on N/N_0. Then do the experiment.

Home experiment

2.2 Slinky as a continuous system. Fasten each end to something fixed. (Tape, string, "eyes" (as in hook and eye), and C clamps may possibly be useful.) A convenient length is about 8 or 10 ft. Don't worry about the sag. Excite the lowest transverse mode in each transverse direction. Measure the frequencies for both modes. Also excite the lowest longitudinal mode and measure its frequency. (There are two good ways to excite a desired mode. One is to constrain the slinky in an appropriate way and then let it go; the other is to hold it near one end and gently shake it at the right frequency until it builds up a decent amplitude, then let go. Use both methods.) Next learn how to excite the second mode, in which the length L is two half-wavelengths. Do this for all three directions, x, y, and z. Measure the frequencies. With some practice you should also be able to excite the third modes.

Now excite the lowest vertical mode and the second longitudinal mode simultaneously. (This can easily be done by a suitable initial constraint.) Look at the system and measure the beat frequency between the longitudinal (second) mode and twice the lowest vertical mode. This is easy once you get the idea and practice for a few minutes. This is a good way to see by inspection that one has an accurate factor of two in frequency in going from the "fundamental" to the "first octave." Similarly you can easily excite the lowest vertical mode and the second horizontal mode simultaneously.

2.3 Null measurements. Read Home Exp. 2.2 (though you need not perform it to work this problem). Suppose you measure slinky frequencies by counting oscillations for about ten seconds and then dividing the number of complete cycles by the time. Suppose that you read the watch to an accuracy of ± 1 sec and that you can estimate a "complete" oscillation to an accuracy of about $\pm\frac{1}{4}$ cycle. The frequency ν_1 of the lowest mode is about 1 cps. The frequency ν_2 of the second mode is about 2 cps.

(*a*) Roughly, what is the fractional (or percent) accuracy that your measurement would give for ν_1? For ν_2? (We want an answer like "$\nu_1 = 1.0 \pm 0.1$, $\nu_2 = 2.0 \pm 0.2$," or whatever it is.)

(*b*) Next, suppose you were to excite both modes simultaneously and measure the beat frequency between $2\nu_1$ and ν_2, as described in Home Exp. 2.2. This could be done by watching for beats for about 10 sec, i.e., about ten cycles of ν_1. Suppose that you could detect, to an accuracy of $\frac{1}{4}$ beat, no beat between $2\nu_1$ and ν_2 in that time. Thus your experimental result would be $\nu_2 - 2\nu_1 = 0$. What is the experimental accuracy? (We want an answer like "$\nu_2 - 2\nu_1 = 0 \pm 0.10\nu_1$", or whatever it is.) What is the accuracy of an estimate of the quantity $\nu_2 - 2\nu_1$ (expressed in the same way) obtained by combining the results of your independent measurements of ν_1 and ν_2 from part (*a*)? Can you see an experimental advantage in the method of counting beats? Explain why you would do much better with that method. Try to generalize this into a statement on "how to make a measurement, if possible."

2.4 "Tone quality" of a slinky. The tone quality of a musical instrument depends on what harmonics are excited. [For example, the even harmonics are missing (or nearly so) from a clarinet; only ν_1, $3\nu_1$, $5\nu_1$, etc., are present.] Locate the center of your slinky (suspended as in Home Exp. 2.2). Excite the slinky by giving it a sudden push at the center with your hand. Try different degrees of abruptness. You should soon be able to see that the even harmonics are always missing and the more impulsive your excitation, the larger the number of (odd) modes you excite. Can you devise a way to excite only the *even* modes? **Home experiment**

Try plucking a guitar or piano string at different places—in the middle or near one end—and see if you can hear a difference in "tone quality."

2.5 Piano as Fourier-analyzing machine—insensitivity of ear to phase. Find a piano. Hold down the damper pedal. Shout "hey" into the region of the strings and sounding board. Listen. Shout "ooh." Try all vowels. The piano strings are picking up (in somewhat distorted form) and then preserving the Fourier analysis of your voice! Notice that the recognizable vowel sound persists for several seconds. What does that tell you about the importance to your ear and brain of the relative phases of the Fourier components that make up the sound? **Home experiment**

2.6 Piano harmonics—equal-temperament scale. Look in the *Handbook of Chemistry and Physics* under "musical scales" for convenient tables of pitches for the three common scales: **Home experiment**

American Standard pitch (A440) equal-temperament chromatic scale
International pitch (A435) equal-temperament chromatic scale
Scientific or Just scale (based on C256, which gives A426.67)

First we shall explain the scientific scale. Let 256 cps equal one unit of frequency, $\nu = 1$. The *harmonics* of this fundamental note are then $\nu = 2, 3, 4$, etc.; the *subharmonics* are defined to be $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$, etc. Middle C on the piano is C256 (if the piano is tuned that way). It is called C_4. (The subscript refers to the octave. The subscript increases by one at each higher octave of C.) Suppose that piano strings perfectly

obeyed the "continuous, perfectly flexible string" dispersion law. Then the mode frequencies of a given string would consist of the harmonic sequence ν_1, $2\nu_1$, $3\nu_1$, etc. The names and frequencies of the first 16 harmonics of string C_4. and also its first two subharmonics would be as follows (we underline C_4 and its octaves):

Name:	F_3	C_3	C_4	C_5	G_5	C_6	E_6	G_6	$B\flat_6$	C_7	D_7	E_7	$F\sharp_7$	G_7	$G\sharp_7$	$B\flat_7$	B_7	C_8
ν:	$\frac{1}{3}$	$\underline{\frac{1}{2}}$	$\underline{1}$	$\underline{2}$	3	$\underline{4}$	5	6	7	$\underline{8}$	9	10	11	12	13	14	15	$\underline{16}$

An octave is always higher by a factor of two in frequency (compare G_6 and G_7). Now let us construct a scale within the single octave between C_4 and C_5 by dividing or multiplying the harmonics and subharmonics of C_4 by appropriate powers of 2. We then get the Scientific or Just diatonic scale in C major (diatonic means we have only the "white notes," none of the "black notes" on the piano keyboard):

Name:	C	D	E	F	G	A	B	C
ν:	1	$\frac{9}{8}$	$\frac{5}{4}$	$\frac{4}{3}$	$\frac{3}{2}$	$\frac{5}{3}$	$\frac{15}{8}$	2

(We sneaked in the A. It is $\frac{5}{4}$F.) The note C is called the *tonic* in this scale.

The smallest musical interval in this diatonic scale is called a *minor second.* The frequency ratio for a minor second is $F/E = C/B = \frac{16}{15} = 1.067$. The next larger ratio is called a major second. There are two kinds of major seconds: $D/C = G/F = B/A = \frac{9}{8} = 1.125$; $E/D = A/G = \frac{10}{9} = 1.111$. There are also two kinds of the next larger ratio, the minor third: $F/D = \frac{32}{27} = 1.185$; $G/E = C/A = \frac{6}{5} = 1.200$. There is only one kind of major third: $E/C = A/F = B/G = \frac{5}{4} = 1.250$. Now comes the difficulty, musically. Suppose that, in the course of composing for a piano tuned to this scale, you suddenly decide that you want to change to a new "key signature," that is, to a diatonic scale with a different note as tonic. For example, you might want to change from C major to D major. You want the same kind of scale, i.e., the same frequency ratios as before. Thus you want the first major second in the new scale, E/D, to be a "D/C type of major second" with ratio 1.125. Unfortunately you can't use the E that you already have, because that gives $E/D = 1.111$. So you need a new string, E′, with $E'/C = (1.125)(D/C) = 1.265$, whereas $E/C = 1.250$. The next note after E′ also demands a new string, called F♯. We make it in the ratio $F\sharp/D = E/C$, so that $F\sharp = (\frac{5}{4})(\frac{9}{8}) = 1.407$. (This is a "black key" on the piano.) Notice that the piano has now acquired a *new kind of minor second:* $F\sharp/F = 1.0555$. As you complete the scale, you must add more and more keys. Then, if you want to play in still other keys, the situation just gets worse and worse. (Try completing the D scale. You have to add the "black key" C♯ to get the note corresponding to B on the C scale. But what other "primed" strings do you need?

The equal-temperament scale gets around all this by making all notes equally spaced on a logarithmic scale. The octave is divided into 12 minor seconds ("half tones"), each of which has the frequency ratio $2^{1/12} = 1.059$. Then all major seconds have the ratio $2^{2/12} = 1.122$; all minor thirds have $2^{3/12}$, etc. None of the intervals are "just" except the octaves, but all intervals are close to the just values in diatonic scales based on any note as tonic.

Try the following experiments:

(i) Steadily hold down the key for, say, $B\flat_6$, so as to lift its damper without sounding the note. Now strike a lower note sharply, hold it for a few seconds, and release the lower key so as to damp the lower note. If you now hear the $B\flat_6$ string sounding, it must have been excited by one of the harmonics present in the mode pattern of the lower vibrating string. Try various lower strings. The note an octave down should work. So should the 12th down, $E\flat$, since $B\flat$ is its third harmonic. So should the C string for which $B\flat_6$ is the 7th harmonic, provided that that harmonic is present in the vibrating string C. Another way to work this experiment is to hit the same lower note, say C_4, while silently holding down different higher notes to see if they are excited. When you have found a note that is excited, try the neighboring note one minor second away. Is it excited?

(ii) This time hold the key of a low note down silently and strike a higher note sharply. If the higher note is one of the overtones of the lower string, you will excite that overtone in the lower string without exciting the lowest mode (the fundamental) of the lower string. Thus you get to hear what the harmonics of the lower string sound like when they are not masked by the loud fundamental.

(iii) Use the method of (ii) to learn what the first 6 or 7 harmonics of C_4 (or a lower C) sound like. Then learn how to hear a particular harmonic in the tone pattern when the lower key is struck in the nomal manner. For example, to learn how to hear the 7th harmonic, $B\flat_6$, when C_4 is struck, hold down C_4 silently and hit $B\flat_6$ sharply. That tells you what $B\flat_6$ sounds like when it is the 7th harmonic of C_4. Then, while your memory is fresh, hit string C_4 and concentrate on picking $B\flat_6$ out of the sound (dominated by the fundamental of C_4). Note that the frequency of this note when it occurs as the 7th harmonic of C_4, i.e., on the C_4 *string*, will not be exactly the same as the frequency of the fundamental note of the $B\flat_6$ string. It will be close enough so that it can be excited, probably, but as soon as the $B\flat_6$ string is damped and the C_4 string has had a few seconds to forget how it was excited, it will oscillate at its own (7th harmonic) frequency, not the exciting frequency. Thus it sounds slightly different from the exciting note. (Of course if the piano is out of tune, it may sound *very* different.) Because of this small frequency difference, you can hear beats as follows:

(iv) Hold down C_4 silently. Hit C_5 sharply. This excites the second mode of the C_4 string. Now, before this dies away, damp the C_5 string and then hit C_5 again, gently, trying to match its loudness to what is left of the second harmonic of C_4. Listen for beats. (This works better on some pianos than on others. It should be done in an otherwise quiet room.)

(v) The lowest two notes on the piano are $A_0$27.5 and $A\sharp_0$29.1. Their beat frequency is thus 1.6 cps, which is easily detectable. Hit both notes together, gently. Once you think you hear beats, let one key up, but not the other. Do the beats go away? (Is the piano in tune?)

2.7 Why an ideal continuous string gives exactly "harmonic" frequency ratios but a beaded string does not. Consider a beaded string of very many beads (say 100) with both ends fixed. We will think of this string as being essentially continuous. Suppose it is oscillating in its lowest mode; then the length L is one half-wavelength of a sine wave. Now consider the second mode: The length L is two half-wavelengths, so the first half of L is one half-wavelength. Now compare the 50 beads in the first half of the string when it is in its second mode, with the entire 100 beads when the string is in its lowest mode. In each case the beads follow a curve that is one half-wavelength of a sine wave. Compare bead 1 (in mode 2) with the average of beads 1 and 2 (in mode 1); compare bead 2 (in mode 2) with beads 3 and 4 (mode 1), etc. Thus, in mode 2, bead 17 has the same amplitude as the average of beads 33 and 34 in mode 1 (if the sine waves have the same amplitude). But in mode 2 the string at bead 17 makes twice as great an angle to the equilibrium axis as does the string at beads 33 and 34 in mode 1 (using the small-angle approximation). Thus the return force per unit displacement on bead 17 is twice that on the two beads 33 and 34. Also, the mass of bead 17 is only half that of the two beads 33 and 34. Thus *the return force per unit displacement per unit mass is four times greater for bead 17 in mode two than it is for the combination of beads 33 and 34 in mode one.* Thus we find, in the "nearly continuous" approximation (implied by our "large number" of beads), $\omega_2 = 2\omega_1$.

This argument does not work if the number of beads becomes small. Explain why not. Then you can see why you get the "harmonic" ratios $\nu_2 = 2\nu_1$, $\nu_3 = 3\nu_1$, etc. in the continuous limit, but not when there are only a few beads, as shown for example in Fig. 2.12.

2.8 How many years does it take to double your money if you invest it at 5.9% interest (per year, compounded annually)? [*Hint:* Consider the equal-tempered scale (Home Exp. 2.6).]

2.9 Complete the "just" diatonic scale for D major which we started in Home Exp. 2.6. We found there that we had to add a new E string, which we called E′. We needed our first "black note," F♯. We will also need another black note, C♯. What about G, F, A, and B? Can we use the ones we have, or do we need G′, F′, A′, and B′?

2.10 Derive Eq. (55), the wave equation for a nonuniform string.

2.11 Obtain the result Eq. (47) for the Fourier coefficients of $F(z)$ plotted in Fig. 2.6.

2.12 Find the mode configurations and frequencies for the first three modes of transverse vibration of a continuous string with tension T_0, mass density ρ_0, and length L, given the boundary conditions that *both ends are free.* (They slide on frictionless rods that pass through massless rings at each end of the string.) Show that the lowest mode has the peculiar property of having infinite wavelength and zero frequency. In this mode, the string translates with uniform velocity. (That includes the possibility of its remaining at rest with arbitrary displacement.)

2.13 Find the three mode configurations and frequencies for transverse vibrations of a uniformly beaded string having three beads and four string segments, given the boundary conditions that both ends are free. (The end string segments have massless rings at their ends and slide on frictionless rods.) Compare the lowest mode with that in Prob. 2.12.

2.14 Consider an LC network of three inductances and four capacitances, arranged as in Fig. 2.15 for $N = 3$, except that *the outer two capacitors are short-circuited.* Find the three modes—current configurations and frequencies. Compare the physical significance of the "peculiar" lowest mode in this problem with that in Prob. 2.13. Compare the boundary conditions with those in Prob. 2.13.

2.15 Consider the piano string which sounds middle C, C256 (Scientific scale). The density of the steel of the string is about 9 gm/cm^3. (This is *not* the linear mass density ρ_0. Why not?) Suppose the string diameter is $\frac{1}{2}$ mm and its length is 100 cm. What is its tension in dynes, in kg-wt, and in pounds? 980 dynes = 1 gm-wt; 354 gm-wt = 1 lb.) *Ans.* $T_0 \approx 47$ kg-wt ≈ 104 lb.

2.16 Find $\psi(z,t)$ for a slinky constrained at $t = 0$ to follow the function $g(z)$ given by Eq. (48). Plot $\psi(z,t_0)$, where $\omega_1 t_0 = \pi/3$. Compare the shape of $\psi(z,t_0)$ with that of $\psi(z,0)$, which is shown in Fig. 2.7.

2.17 Compare the tension in a steel guitar string with that in a catgut string of the same length, diameter, and pitch (of lowest mode). The density of steel is about 9 gm/cm^3; the density of catgut is not much more than 1 gm/cm^3. Are the string diameters actually the same for steel-string guitars as for catgut-string guitars? Look at guitars and find out. Once you have looked and estimated the ratio of diameters, recalculate the ratio of tensions in the two cases.

2.18 Derive the classical wave equation [Eq. (14)] in the following way: Start with the exact Eq. (62). Now go to the continuous approximation. Replace subscript n by location z, taking into account the bead separation is length a. Use the Taylor's series expansions of the right side of Eq. (62). *Include one more term than is necessary to obtain the classical wave equation.* Give a criterion for neglecting this and higher-order terms.

2.19 Show that by taking Eq. (71) as the solution to the equation of motion for transverse vibrations of the beaded string, Eq. (65), one gets the dispersion relation Eq. (70). Show that this fact is independent of the choices for the constants A, B, and k, which depend only on initial conditions and boundary conditions.

2.20 Use Eqs. (73) and (70) to obtain the frequency ratios shown in Fig. 2.12 for $N = 5$.

2.21 Find the mode configurations and frequencies for transverse oscillations of a beaded string of 5 beads with one end fixed and one free. Plot the five corresponding points on the dispersion relation $\omega(k)$ as in Fig. 2.13.

2.22 By inspection of Fig. 2.13 and a sketch of the system, show an easy way to add 6 more points to Fig. 2.13, so that it gives the modes for a beaded string of 11 beads with both ends fixed.

2.23 Show that Eqs. (73) and (74) give the same answers for the frequencies for $N = 1$ and $N = 2$ as those we obtained in Sec. 1.2 and Sec. 1.4.

2.24 Sketch the configurations of the five modes of a 5-beaded string corresponding to Eqs. (78) to (80).

2.25 Plot the dispersion relation for the system shown in Fig. 2.15.

2.26 Show that, for the system of coupled pendulums shown in Fig. 2.16, the equation of motion for the nth pendulum bob is given (for small oscillations) by

$$\frac{d^2\psi_n}{dt^2} = -\frac{g}{l}\,\psi_n + \frac{aK}{M}\left(\frac{\psi_{n+1} - \psi_n}{a}\right) - \frac{aK}{M}\left(\frac{\psi_n - \psi_{n-1}}{a}\right)$$

Show that the general solution for a mode, without regard to boundary conditions, is

$$\psi_n(t) = \cos(\omega t + \varphi)[A \sin nka + B \cos nka].$$

Show that the dispersion relation is

$$\omega^2 = \frac{g}{l} + \frac{4K}{M}\sin^2\frac{ka}{2}$$

Show that, for the boundary conditions shown in Fig. 2.16 (i.e., no springs coupling the end bobs to the wall), the above solution reduces to

$$\psi_n(t) = \cos(\omega t + \varphi)B \cos nka,$$

with the nth bob located at $z = (n - \frac{1}{2})a$. Show that the lowest mode has $k = 0$. Sketch its configuration. What would be the behavior of the system in this configuration if the gravitational constant were gradually reduced to zero? Sketch the configurations for $N = 3$ and give the frequencies for the three modes.

2.27 Find the system of coupled capacitances and inductances which is "analogous" to the system of coupled pendulums of Fig. 2.16, in the sense that the equation of motion for the nth inductance has the same form as the equation of motion for the nth pendulum bob, which you found in Prob. 2.26. Find the dispersion relation.

2.28 Go to the continuous limit in the coupled-pendulum problem (Prob. 2.26). Show that the equation of motion becomes a wave equation of the form

$$\frac{\partial^2\psi}{\partial t^2} = -\omega_0{}^2\psi + v_0{}^2\frac{\partial^2\psi}{\partial z^2}.$$

2.29 Prove each of the following numbered statements using two methods: (*a*) the "physical" method that makes use of the normal modes of a continuous string with appropriate boundary conditions, and (*b*) the method of Fourier analysis of a periodic function of z.

(i) Any (reasonable) function $f(z)$ defined between $z = 0$ and $z = L$ and having value zero at $z = 0$ and slope zero at $z = L$ can be expanded in a Fourier series of the form

$$f(z) = \sum_n A_n \sin nk_1 z; \qquad n = 1, 3, 5, 7, \ldots; \qquad k_1 L = \frac{\pi}{2}.$$

(*Note:* In using the method of Fourier analysis you must first construct a periodic function from $f(z)$ so that you can use the formulas of Fourier analysis.)

(ii) Any (reasonable) function $f(z)$ defined between $z = 0$ and $z = L$ and having zero slope at $z = 0$ and zero slope at $z = L$ can be expanded in a Fourier series of the form

$$f(z) = B_0 + \sum_n B_n \cos nk_1 z; \qquad n = 1, 2, 3, 4, \ldots; \qquad k_1 L = \pi.$$

(iii) Any (reasonable) function $f(z)$ defined between $z = 0$ and $z = L$ and having slope zero at $z = 0$ and value zero at $z = L$ can be expanded in a Fourier series of the form

$$f(z) = \sum_n B_n \cos nk_1 z; \qquad n = 1, 3, 5, 7, \ldots; \qquad k_1 L = \frac{\pi}{2}.$$

2.30 Fourier analysis of a periodically repeated square pulse. When you clap your hands periodically, the resulting air pressure at your ear can be approximated by a periodically repeated square pulse. Let $F(t)$ represent the gauge pressure at your ear. Take $F(t)$ to be $+1$ unit for the short time interval Δt, and zero before and after that interval. This "square pulse" of unit height and width Δt [on a plot of $F(t)$ versus t] is repeated periodically at time intervals of length T_1. The short interval Δt gives the duration of each clapping sound. The period T_1 is the time between successive claps. The frequency $\nu_1 = T_1^{-1}$ is the clapping frequency. You are to Fourier-analyze $F(t)$.

(*a*) Show that you can choose the time origin so that only cosines of $n\omega_1 t$ appear, i.e., so that

$$F(t) = B_0 + \sum_{n=1}^{\infty} B_n \cos n\omega_1 t.$$

(*b*) Show that $B_0 = \Delta t / T_1$, which is just the fractional "on" time. Show that

$$B_n = \frac{2}{n\pi} \sin (n\pi\nu_1 \Delta t), \qquad \text{for } n = 1, 2, \ldots.$$

(*c*) Show that for $\Delta t \ll T_1$ the "fundamental" tone ν_1 and the low harmonics $2\nu_1$, $3\nu_1$, $4\nu_1$, etc. all have essentially the same value for their Fourier amplitudes B_n.

(*d*) Sketch B_n versus $n\nu_1$, going up to sufficiently high n so that B_n has gone through zero two or three times.

(*e*) Show from part (*d*) that the "most important" frequencies (i.e., those with relatively large values for B_n) go from the fundamental, ν_1, to a frequency of the order of $1/\Delta t$. Thus we may call $1/\Delta t$ by the name $\nu_{\max}$. Actually, of course, there is no maximum frequency since the Fourier series extends to $n = \infty$. However, the most important frequencies lie between zero and $\nu_{\max}$. The "frequency band" of dominant frequencies has "bandwidth" equal to about $\nu_{\max} = 1/\Delta t$. The important frequencies are thus

$$\nu = 0,\ \nu_1,\ 2\nu_1,\ 3\nu_1,\ 4\nu_1,\ \ldots\ \nu_{\max} = \frac{1}{\Delta t}$$

The bandwidth of dominant frequencies can be given the name $\Delta\nu$. Then your result can be written

$$\Delta\nu\,\Delta t \approx 1.$$

This is a very important relation. It holds not only for our assumed $F(t)$, a repeated "square" pulse of width Δt, but for any pulse shape that can be characterized as being zero most of the time and nonzero for a time of duration about Δt. If (as in our example) the pulse is repeated at intervals T_1, then the dominant frequencies are zero, ν_1, $2\nu_1$, $3\nu_1$, etc. up to about $1/\Delta t$. If the pulse is not repeated but only occurs once, then it turns out (as we shall show in Chap. 6) that the "Fourier spectrum" of important frequencies still occupies the frequency band extending from zero to about $1/\Delta t$, but it is a continuous spectrum, including all frequencies in the band rather than just the fundamental, ν_1, and its harmonics.

The present problem can help you to understand the frequency spectrum of the electromagnetic radiation called *synchrotron radiation* which is emitted by a relativistic electron undergoing uniform circular motion. It can be shown (Chap. 7) that a nonrelativistic electron undergoing uniform circular motion at frequency ν_1 emits electromagnetic radiation having the single frequency ν_1. That is because the electric field in the radiation for nonrelativistic electron velocities is just proportional to the component of acceleration of the charge perpendicular to the line of sight from the charge to the observer. For circular motion this projected acceleration is simple harmonic motion. Therefore for a nonrelativistic electron the radiated field is proportional to the cosine (or sine) of $\omega_1 t$. For a relativistic electron, the radiated field does not have the time dependence $\cos\omega_1 t$. Instead the radiation intensity is very strongly concentrated in direction, along the instantaneous direction of the velocity of the charge. When the electron is headed directly toward the observer, it is emitting radiation that will later be detected by the observer. At other times, the radiation it is emitting will not reach the observer. Thus the electric field measured by the observer is strong for a short time interval Δt once every period T_1, and nearly zero at other times. Thus the frequency spectrum observed consists of $\nu_1 = 1/T_1$ and its harmonics $2\nu_1$, $3\nu_1$, etc., up to a maximum (important) frequency of about $1/\Delta t$. Show that the time interval Δt is given roughly by $\Delta t/T_1 \approx \Delta\theta/2\pi$, where $\Delta\theta$ is the "angular full-width" of the radiation pattern.

2.31 Sawtooth shallow-water standing waves. Shallow-water waves are waves in which the amplitude of motion of the water at the bottom of the pan, lake, or ocean is comparable in magnitude with that at the surface. The sloshing mode (Home Exp. 1.24) is a shallow-water wave. Show this experimentally by stirring some coffee grounds into the water so that some grounds are near the bottom. Excite the sloshing mode (the one where the surface remains essentially flat) and look at the motion of coffee grounds at the bottom and at the surface near the center of the pan. Look also near the end walls.

Now consider the following idealized *sawtooth shallow-water standing wave.* Suppose you have two independent pans of the same shape with water of the same equilibrium depth, h, undergoing oscillation in the sloshing mode. The pans are adjacent so that, if there were no partition, the two would form one long pan along the horizontal oscillation direction. Suppose the phases of the oscillations are such that water in one pan always moves horizontally opposite to that in the other, causing the water to pile up to its maximum height in both pans at the same time at the walls separating the two. Now imagine that you remove the walls separating the two pans. The water at the boundary surface had no horizontal motion when the walls were in place. It still has none because of the symmetry of the motion of the two bodies of water (now joined to form one long body). The motion should continue unchanged! We can continue joining pans if we wish. We have here a standing wave with a sawtooth shape. Let us approximate that shape by a sine wave. Then we see that *one pan length equals one half-wavelength.* (*Note:* If you Fourier-analyze this periodic function of z, the first and most dominant term in the Fourier expansion will be the one that we are using to approximate the sawtooth.) Use that approximation in the formula that gives the frequency of the sloshing mode (see Home Exp. 1.24). Show that one thus obtains

$$\lambda\nu = \frac{2\sqrt{3}}{\pi}\sqrt{gh} = 1.10\sqrt{gh}.$$

We see that *these waves are nondispersive.* (*Note:* It turns out that the exact dispersion relation for sinusoidal shallow-water waves is $\lambda\nu = \sqrt{gh}$. Our sawtooth approximation gives a propagation velocity 10% too high.)

For *deep-water waves* (waves where the equilibrium water depth is large compared with the wavelength), the wave amplitude falls off exponentially with depth below the surface, with a decrease in amplitude by a factor of $e = 2.718\ldots$ for every increase in depth by an amount $\lambda\!\!\!^{-} \equiv \lambda/2\pi$, where $\lambda\!\!\!^{-}$ (pronounced "lambda bar") is called the reduced wavelength. To a crude approximation we can say that a deep-water wave is something like a shallow-water wave from the surface down to an effective depth $h = \lambda\!\!\!^{-}$, since in that region the amplitude is relatively large and roughly constant, whereas for depths much greater than $\lambda\!\!\!^{-}$ the amplitude is very small. Thus we guess that the dispersion relation for deep-water waves may be obtained from that for shallow-water waves by replacing the equilibrium depth h for shallow-water waves by the mean amplitude attenuation length $\lambda\!\!\!^{-}$ for deep-water waves. That conjecture turns out to be correct, as we shall show in Chap. 7. Thus the dispersion law for deep-water waves is given by $\lambda\nu = \sqrt{g\lambda\!\!\!^{-}}$.

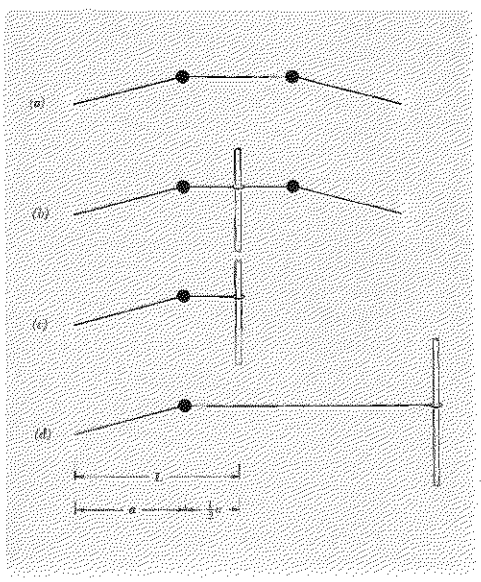
2.32 Fourier analysis of symmetrical sawtooth. By a symmetrical sawtooth we mean one where the front and back edges of each tooth have the same slope. Let $z = 0$ occur at one of the crests (tooth points). Show that the periodic sawtooth $f(z)$ has a Fourier series given by

$$f(z) = 0.82A[\cos k_1 z + \tfrac{1}{4}\cos 2k_1 z + \tfrac{1}{9}\cos 3k_1 z + \cdots],$$

where $k_1 = 2\pi/\lambda_1$, where λ_1 is the tooth length (from one crest to the next), and where A is the tooth amplitude, i.e., $2A$ is the vertical distance from the minimum (at a trough) to the maximum (at the crest). Thus we see that the contribution from the nth term has amplitude proportional to $1/n^2$. This tells you something about how good our approximation was in Prob. 2.31, where we approximated the sawtooth by its first Fourier component to obtain the dispersion relation $\lambda\nu = 1.10\sqrt{gh}$.

Home experiment

2.33 Surface tension modes. Circular surface-tension standing waves can be clearly demonstrated as follows: Fill a paper or styrofoam cup to the brim and then add a little more, so that the water bulges above the top of the cup. Tap the cup gently. Watch! To see the waves easily look at the reflection of the sky. Alternatively, use a small bright light held several feet above the surface, and look at the pattern on the bottom caused by the lenslike effect of the waves. To see that surface tension is involved, try adding a little detergent to the water.



Problem 2.34

2.34 Boundary conditions at free end of string. Consider the four different systems shown in the figure.

(i) Show that all four systems have the same frequency in the modes shown.

(ii) Suppose you want to use the same formula for the constraint on wavenumber in cases (c) and (d) as that which is used in case (a). Then show that L in your formula must equal $(\frac{3}{2})a$ in those cases. Give the formula.

2.35 A flexible string of length L is stretched with equilibrium tension T between fixed supports. Its mass per unit length is ρ, so that its total mass is $M = \rho L$. The string is set into vibration with a hammer blow which impulsively imparts a transverse velocity v_0 to a small segment of length a at the center. Evaluate the amplitudes of the lowest three harmonics excited.